Long-Read Soil Metagenomic Assembly and Differential Abundance Analysis Across Plant Shoot Biomass Gradients

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A thesis submitted for the degree of

Master of Science (Advanced) in Quantitative Biology and Bioinformatics

The Research School of Biology

Australian National University

October 2025

Word count: 10,297

Declaration of Original Work

I hereby declare that this thesis is my own work and that, except where due acknowledgment is made in the text.

The research presented in this thesis was conducted under the supervision of Professor Justin Borevitz. Samples and experimental data were collected by collaborators as acknowledged in Section 2.1.1 (Methods). All bioinformatic analyses, data interpretation, and writing of this thesis are my original work.

Heyue Wu

26 October 2025

Acknowledgement

I am deeply grateful to my supervisor, Professor Justin Borevitz, for his invaluable guidance throughout this research. Your insights illuminated the path forward at countless moments when I felt lost, gently steering me toward clarity. Your trust, encouragement, and affirmation gave me tremendous confidence and motivation to pursue this work. Thank you for believing in me.

I would like to thank all my colleagues in the Borevitz Lab. You are exceptional scientists and wonderful people. I have learned so much from each of you. Thank you for your inspiration and support throughout this journey.

Special thanks to Viraj Kolhapuri and Dmitry Grishin, who have been with me from the beginning of this project. We supported each other through the early challenges, and your friendship and collaboration have continued to this day. I am grateful for your companionship.

I am indebted to Alek Meade, Ashley Jones, and Christopher Bradley for patiently answering my many bioinformatics questions and sharing their expertise. Your generosity with your time and knowledge made this work possible.

I extend my sincere gratitude to my examiners, Professor Benjamin Schwessinger and Professor Robert Lanfear, for their thoughtful feedback and guidance. Every comment and suggestion you provided has been invaluable, and I deeply appreciate the time and care you invested in reviewing this work.

Finally, and most importantly, I thank my family, my partner, and my friends. You have filled my life with joy and beauty, and your love and support sustained me through this endeavour. This achievement is as much yours as it is mine.

Abstract

Soil microbiomes play critical roles in carbon cycling and storage, yet linking specific microbial genomes to their ecological functions remains challenging. Traditional short-read metagenomic approaches produce fragmented assemblies that limit confident taxonomic assignment and functional annotation. This thesis applies Oxford Nanopore long-read sequencing to reconstruct high-quality, near-complete microbial genomes from soil communities and quantify how microbial composition and function shift across environmental gradients.

We sequenced soil metagenomes from three complementary experiments spanning temporal dynamics, plant species effects, and soil type contrasts. Using a unified co-assembly strategy with nanoMDBG, we assembled 556,967 contigs totalling 7.69 Gbp. The assembly recovered 86 contigs exceeding 1 Mb (longest: 4.52 Mb) and 923 circular replicons representing complete genetic elements. We achieved 20% higher N50 and an 8-fold enrichment in megabase-scale contigs compared to the global short-read SMAG catalogue, with 2,311 complete rRNA operons enabling confident taxonomic placement.

Differential abundance analysis of plant culture experiments identified 194 contigs whose abundances varied systematically with shoot biomass, revealing a clear copiotrophic-oligotrophic gradient. High-biomass rhizospheres were dominated by fast-growing Actinobacteriota (55% of responsive contigs, primarily Mycobacteriales including Pseudonocardia, Geodermatophilus, and Actinoplanes) and Proteobacteria (23%, including Massilia). In contrast, low-biomass soils harboured oligotrophic specialists including Acidobacteriota (14%), Gemmatimonadota (17%), and distinct Proteobacteria lineages adapted to nutrient limitation. Notably, 32% of high-biomass contigs remained taxonomically unclassified, highlighting substantial genomic novelty.

These findings demonstrate that long-read metagenomics enables genome-resolved investigation of soil microbiomes, connecting microbial identity to metabolic potential and

ecological function. The dominance of copiotrophic taxa in productive rhizospheres, coupled with their rapid turnover, likely accelerates conversion of root-derived carbon into stable microbial necromass, which is a key mechanism in the living soil pathway for long-term carbon storage. This genome-resolved approach provides a foundation for manipulating soil microbiomes to enhance carbon sequestration and sustainable agriculture.

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List of Abbreviations

AS	Alignment Score (in sequence mapping)
auN	Area Under the Nx curve (assembly contiguity metric)
BAM	Binary Alignment Map (compressed sequence alignment format)
BH	Benjamini-Hochberg (multiple testing correction method)
BGC	Biosynthetic Gene Cluster
bp	base pairs
CLR	Centred Log-Ratio (transformation for compositional data)
CoDA	Compositional Data Analysis
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats
CSIRO	Commonwealth Scientific and Industrial Research Organisation
^{13}C	Carbon-13 (stable isotope)
DA	Differential Abundance
DNA	Deoxyribonucleic Acid
EC	Electrical Conductivity
FASTQ	Text-based format for storing nucleotide sequences and quality scores
FDR	False Discovery Rate
Gbp	Gigabase pairs (one billion base pairs)
GC	Guanine-Cytosine (content)
GTDB	Genome Taxonomy Database

HiFi	High-Fidelity (PacBio sequencing technology)
IQR	Interquartile Range
kb	kilobase (one thousand base pairs)
log2FC	log2 Fold Change (effect size in differential abundance)
MAG	Metagenome-Assembled Genome
MAOM	Mineral-Associated Organic Matter
MAPQ	Mapping Quality (probability that read is mapped correctly)
Mb	Megabase (one million base pairs)
MDBG	Minimizer-space de Bruijn Graph (assembly algorithm)
N50	Length of the shortest contig at which 50% of assembly is contained in contigs of that length or longer
ONT	Oxford Nanopore Technologies
PC1	Principal Component 1 (first principal component)
PCA	Principal Component Analysis
PERMANOVA	Permutational Multivariate Analysis of Variance
POM	Particulate Organic Matter
Q	Quality score (Phred quality score for sequencing accuracy)
QL	Quasi-Likelihood (statistical framework in edgeR)
RNA	Ribonucleic Acid
rRNA	ribosomal RNA
SAM	Sequence Alignment Map (text format for sequence alignments)
SMAG	Soil Metagenome-Assembled Genome (global catalogue)

SOC	Soil Organic Carbon
SOM	Soil Organic Matter
sup	super-accurate (basecalling mode in Dorado)
TMM	Trimmed Mean of M-values (normalization method)
TMMwsp	TMM with singleton pairing (for sparse data)

Chapter 1

Introduction

1.1 Soil metagenomics and soil carbon cycling

The soil microbiome is one of the most biodiverse and functionally important ecosystems on Earth. Soil microbes play key roles in nutrient cycling, plant growth, and global biogeochemical processes. They decompose plant litter and metabolize root exudates, respiring CO_2 while transforming organic matter into more stable soil organic matter (SOM). Microbes release carbon and nutrients through catabolic breakdown of organic matter. Conversely, anabolic processes incorporate carbon into microbial biomass, which—after cell death—contributes to long-term soil carbon storage. The accumulation of microbial necromass (dead cells and their remnants) through repeated biomass turnover is now recognized as a dominant mechanism of carbon stabilization. This process, known as the "microbial carbon pump," converts easily degradable plant compounds (e.g., lignin) into more chemically stable microbial products (e.g., amino sugars). These compounds resist further decay, helping carbon persist in the soil. In this way, microbes act not only as decomposers but also as key engineers of soil carbon storage (Overy et al., 2021).

However, a major challenge in soil microbiome research is linking specific microbes ("who is there") to their ecological functions ("who does what"). Traditional marker gene sequencing only captures short segments of DNA, providing limited taxonomic information. This is particularly limiting in soils, where more than 99% of microbes are uncultivated and highly diverse. As a result, many microbial species and their genes remain unclassified, making it difficult to study key processes like how microbes help form persistent SOM or how rare microbes contribute to ecosystem function (Overy et al., 2021).

Soil metagenomics offers a solution. This approach involves extracting and sequencing DNA directly from soil, without the need to culture organisms in the lab. This method captures both the structure (which microbes are present and in what amounts) and the functional potential

(what genes and metabolic pathways they carry) of microbial communities. It has uncovered an extraordinary range of previously unknown microbes, which are often referred to as "microbial dark matter" and were not accessible with older methods (Ma et al., 2023; Overy et al., 2021).

However, there is an important limitation. Soil microbes often exist in a dormant or low-activity state, surviving on minimal resources. This means that metagenomic data reflect potential capabilities rather than current activity (Hobbie & Hobbie, 2013). To address this, long-read, genome-resolved metagenomics is becoming essential for connecting microbial identity to function, especially in studies of soil carbon dynamics (Overy et al., 2021).

Understanding how microbial community structure and function influence whether carbon is released or retained is essential for managing soil carbon storage. Metagenomic tools are now being used to track carbon movement through the soil–microbe system and to connect agricultural practices with changes in microbial genes and soil carbon outcomes (Overy et al., 2021). This highlights why studying microbial communities through metagenomics is key to improving soil carbon sequestration strategies.

1.2 Living soil pathway vs. decomposition pathway

Understanding which microbes are present and what functions they encode provides the foundation for studying how microbial communities influence soil carbon dynamics, which follow two main pathways for carbon input and stabilization.

Soil carbon enters and stabilizes through two main pathways (Figure 1.1). The decomposition pathway begins with the breakdown of dead plant litter. This process mainly produces particulate organic matter (POM) and releases CO_2. Carbon in this pathway is only weakly stabilized unless it becomes physically protected inside soil aggregates. In contrast, the living soil pathway directs carbon through interactions between plant roots and soil microbes. Here, carbon is incorporated into microbial biomass and residues, which can bind to minerals and form stable, long-lasting mineral-associated organic matter (MAOM) (Buss et al., 2025; Prescott & Vesterdal, 2021). These pathways differ in both carbon source and the type of soil

organic matter they tend to form. The living pathway, fuelled by root inputs, favours the formation of dense, stable MAOM. The decomposition pathway, driven by dead litter, mostly produces lighter, less stable POM (Figure 1.1). Understanding how much each pathway contributes to soil carbon is key to predicting carbon outcomes under different land management strategies.

Understanding of soil carbon dynamics has shifted from decomposition-centered models to a "living soil" perspective. Classical models (e.g., RothC, Century) treated soil as a passive repository where litter decomposed at rates governed by substrate chemistry, with above- and below-ground inputs having equivalent potential to form stable SOC (Berhongaray et al., 2019). Management strategies consequently focused on slowing decomposition through recalcitrant inputs or reduced disturbance. This framework has now been challenged by evidence that microbial processing and mineral interactions, rather than inherent substrate chemistry, primarily control carbon persistence.

More recent research has reframed this understanding. Soil is now recognized as a dynamic, living system where interactions between plants and microbes are central to carbon stabilization. In this updated perspective, both the pathway and quality of carbon inputs matter. Carbon that enters soil through living roots, root exudates, and microbial uptake is far more likely to form stable soil organic matter than carbon from bulk litter decomposition (Buss et al., 2025).

Empirical evidence strongly supports this shift toward a living soil framework. Berhongaray et al. (2019) showed that below-ground inputs contribute much more to SOC formation. In a bioenergy poplar plantation, root-derived carbon led to the formation of nearly twice as much new SOC as shoot litter (29 g C m^{-2} from roots compared to 16 g C m^{-2} from leaves) even though root inputs made up only about 20% of the total biomass.

The difference in efficiency was striking. About 76% of the root-derived carbon was stabilized as SOC, compared to just 9% of the carbon from shoot litter. Most of the litter carbon was broken down and released as CO_2 (Berhongaray et al., 2019). This contrast reflects the different fates of carbon along each pathway: root-derived carbon is quickly taken

up by microbes and incorporated into dense MAOM, while litter-derived carbon tends to remain as less-stable POM or is lost through respiration (Figure 1.1).

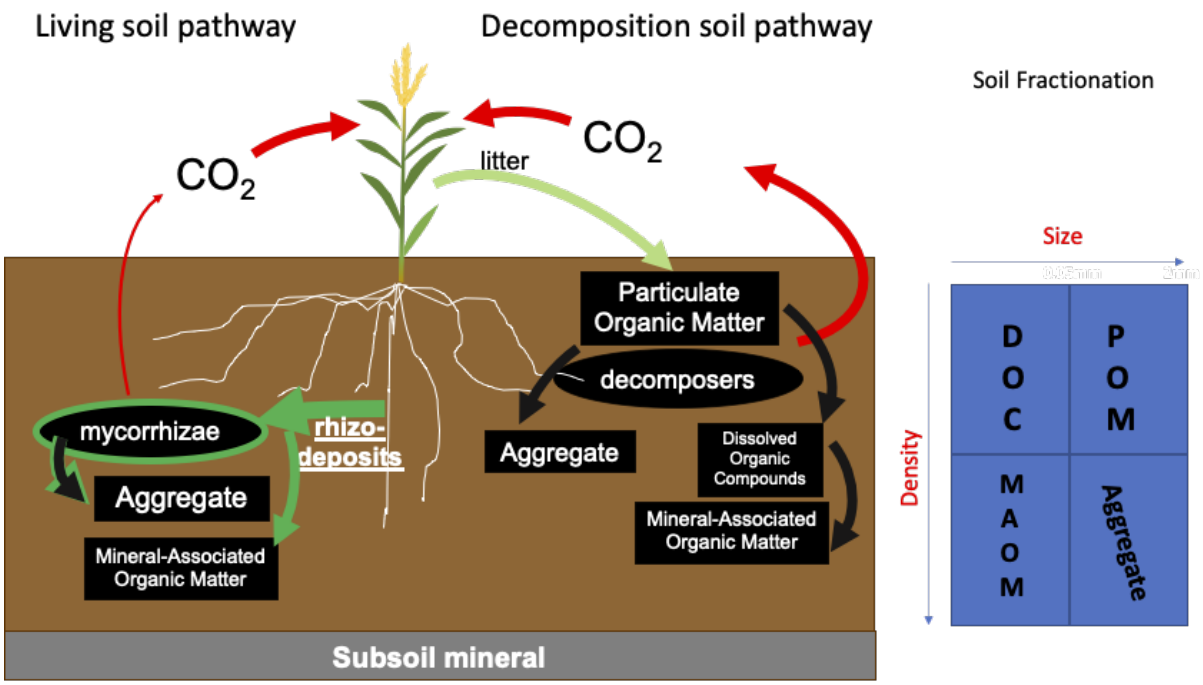


Figure 1.1 Living soil versus decomposition pathways of carbon stabilization. The living soil pathway (left) channels carbon through root exudates, mycorrhizal associations, and rhizosphere interactions directly into stable forms: mineral-associated organic matter (MAOM) and aggregates. The decomposition pathway (right) processes aboveground litter through microbial decomposers, forming primarily particulate organic matter (POM) with greater CO_2 losses. The soil fractionation scheme (right panel) illustrates how organic matter fractions differ by size and density: MAOM represents the most stable, mineral-bound carbon, while POM is less dense and more readily decomposable. DOC dissolved organic compounds. From Buss et al. (2025).

The rhizosphere (the narrow zone surrounding living roots) is a key site in this process. It acts as a biological hotspot where microbial growth, aggregation, and carbon protection are actively stimulated. Because the living soil pathway operates at this plant-microbe interface, the strength of root-microbe interactions can be assessed through the relative abundance of microbial versus plant DNA in soil samples. Rhizosphere samples obtained by washing root-adhering soil are expected to contain both microbial and plant DNA, with the microbe:plant ratio serving as a proxy for rhizosphere influence and the intensity of root-microbe coupling. Higher microbial signal relative to plant signal indicates active microbial colonization and growth stimulated by root exudation, consistent with enhanced microbial biomass turnover and MAOM formation characteristic of the living soil pathway. This ratio can thus serve as

both a quality indicator (confirming samples represent the rhizosphere zone) and an ecological metric that may correlate with carbon stabilization potential across different plant species, soil types, and management practices. As microbial cells die, their necromass contributes to stable carbon pools that can persist for decades or even centuries (Buss et al., 2024; Wang & Kuzyakov, 2024).

These two frameworks lead to very different approaches to soil carbon management. Decomposition-centric approaches focus on slowing carbon loss by adding resistant amendments, such as biochar or high-lignin residues, or by reducing soil disturbance. In contrast, living soil strategies aim to build stable carbon by promoting biological processes. These include encouraging root growth and microbial biomass through deep-rooted crops, cover cropping, and practices that increase below-ground carbon inputs and microbial necromass production (Wang & Kuzyakov, 2024).

One emerging approach, known as rhizosphere engineering, explicitly targets the living components of the system to enhance carbon sequestration. It focuses on increasing below-ground plant productivity, stimulating microbial biomass, protecting organic matter through soil aggregation, and minimizing carbon losses (Wang & Kuzyakov, 2024). By promoting the living soil pathway through continuous root growth, active mycorrhizal networks, and diverse rhizosphere communities, these strategies offer not only climate mitigation potential but also co-benefits such as improved soil structure, water retention, and fertility.

Importantly, these perspectives are not mutually exclusive. Decomposition and stabilization happen simultaneously through microbial activity (Buss et al., 2025). Microbes break down plant litter, releasing CO_2, but they also transform plant-derived carbon into biomass and secretions, some of which become stabilized in soil.

A key challenge is to better understand these dynamics, specifically, to determine which microbial activities or specific taxa are associated with carbon loss versus carbon stabilization under varying environmental conditions (Overy et al., 2021). At broader ecosystem scales, microbial life-history strategies, such as copiotrophic (fast-growing) versus oligotrophic (slow-growing) lifestyles, likely influence whether carbon is respired or stored. This

highlights the need to incorporate microbial traits into predictive ecosystem and carbon cycle models (Trivedi et al., 2013).

Framed within the living soil perspective, this study uses advanced metagenomic tools to connect microbial diversity and function with soil carbon outcomes under different experimental treatments. A key focus is resolving microbial functional contributions at fine scales to better understand how microbes influence carbon retention or loss.

1.3 Soil MAG catalogues and short-read metagenomics approaches

In recent years, large-scale genome catalogues have transformed our understanding of soil microbiome diversity. A landmark example is the Soil Metagenome-Assembled Genome (SMAG) catalogue compiled by Ma et al. (2023). By aggregating 3,304 soil metagenomic datasets from around the world, the researchers reconstructed 40,039 metagenome-assembled genomes (MAGs), which are draft genomes binned from assembled DNA contigs. These MAGs were grouped into approximately 21,000 species-level genome bins (SGBs). Remarkably, about 78% of these represented previously unknown species with no close cultured relatives (Ma et al., 2023). The SMAG catalogue revealed an immense diversity of novel soil microbes spanning 88 bacterial phyla and 11 archaeal phyla, dramatically expanding the microbial Tree of Life.

Beyond taxonomic discoveries, these genomes also contained substantial functional potential. The catalogue identified over 43,169 biosynthetic gene clusters linked to secondary metabolite production, as well as more than 8,500 CRISPR-Cas systems from previously uncharacterized lineages (Ma et al., 2023). Together, these resources provide a powerful genomic reference for soil microbiome research. The SMAG catalogue improves the accuracy of read mapping, gene discovery, and functional annotation in future metagenomic studies.

The success of projects like the SMAG catalogue builds on well-established short-read sequencing workflows. In the study, community DNA is extracted from soil and sequenced using high-throughput platforms such as Illumina, generating millions of short reads (typically around 150 base pairs in length). These reads are then assembled de novo into

longer contiguous sequences (contigs), using algorithms that piece together overlapping sequences. This step is particularly challenging in soil, where microbial communities can contain thousands of different species. Next, binning algorithms group contigs that likely belong to the same genome based on features such as nucleotide composition (e.g., GC content, tetranucleotide frequency) and read coverage patterns across multiple samples. Contigs with similar characteristics are clustered together to reconstruct metagenome-assembled genomes (MAGs), which represent draft genomes of individual microbial populations within the community. Once assembled, MAGs are taxonomically classified and functionally annotated to predict the metabolic capabilities of uncultured microbes. This short-read pipeline (from sequencing to assembly, binning, and analysis) has become standard in metagenomics. It has enabled major discoveries, including the identification of novel microbial lineages and the expansion of known metabolic pathways in environmental microbes (Ma et al., 2023).

Despite these achievements, short-read metagenomics faces important limitations in complex environments like soil. One major challenge is genome fragmentation. While Illumina reads are highly accurate, their short length makes it difficult to assemble across repetitive or highly conserved regions. Regions such as rRNA operons, transposons, and gene duplications often cannot be resolved, leading to breaks in genome assemblies (Zhang et al., 2023). This issue was evident in the SMAG study, where fewer than 10% of the reconstructed genomes met the criteria for “high-quality” MAGs (defined as ⩾90% complete). Most MAGs were incomplete, and many lacked full rRNA operons even when nearly complete in other respects (Ma et al., 2023). Ribosomal RNA genes are especially problematic. Because they contain near-identical repeat regions, they often cause assembly algorithms to break contigs at these points. As a result, even high-quality MAGs may be missing these essential markers, limiting accurate taxonomic classification and downstream phylogenetic analyses.

A second major limitation of short-read metagenomics lies in the binning process, which can introduce artifacts and obscure strain-level variation. MAGs built from short-read contigs often represent composite genomes, combining sequences from multiple closely related

strains. When multiple strains of a species are present in the same sample, binning may lump them together, if they are too similar to distinguish or fragment a single genome across multiple bins due to uneven read coverage (Benoit et al., 2024). Either way, the result is a loss of true strain-level resolution. In the SMAG study, this issue was evident in the form of intra-MAG strain heterogeneity. High-quality MAGs showed a median of ~7.1% polymorphic sites, suggesting that each MAG often represented a consensus sequence from several distinct strains rather than a single clonal genome (Ma et al., 2023). Across the entire catalogue, more than 582 million single-nucleotide variants (SNVs) were detected within species-level genome bins, highlighting the extent of hidden intra-species genomic diversity that consensus MAGs fail to capture. This micro-heterogeneity can have real functional consequences. For instance, if one strain carries a gene for antibiotic resistance or specialized metabolism and another does not, a composite MAG may either include or omit the gene inconsistently. This blurs the true functional potential of the community. Additionally, low-abundance strains may not assemble well or may be filtered out during binning as "noise." As a result, rare but potentially important strain variants are often missed entirely (Van Goethem et al., 2021).

A third limitation of short-read metagenomics is its inability to resolve structurally complex and repetitive genomic regions. Elements such as large biosynthetic gene clusters (BGC, often >50 kbp), CRISPR arrays, and plasmids are frequently fragmented in short-read assemblies due to their internal repeats and complex structures (Ma et al., 2023). In the SMAG study, most biosynthetic gene clusters were recovered only as partial fragments, with the researchers noting that short-read sequencing significantly limited the reconstruction of full-length sequences. Mobile genetic elements, including plasmids and phages, pose additional challenges. Because they often exist in multiple copies or share repeated sequences with chromosomal DNA, they are prone to mis-assembly or mis-binning. This leads to errors in assigning genes to the correct genome. These assembly gaps make it difficult to link taxonomic identity with ecological function. If a MAG is incomplete or lacks key genes, it's challenging to determine what roles that organism plays in the ecosystem (Ma et al., 2023). Fragmentation can also result in important genes being detected in isolation, without clear information on which microbe carries them, which will obscure crucial microbe-trait

relationships and limiting functional interpretation.

While short-read metagenomic approaches have significantly expanded our understanding of soil microbiome diversity, they inherently sacrifice resolution. These methods provide a broad overview of community gene content and population-level genomes, but at the cost of fragmented assemblies and blended strain-level information. The consensus nature of MAGs is generally sufficient for large-scale surveys of taxonomic diversity and core metabolic functions. However, it can be misleading in studies that aim to resolve fine-scale ecological interactions or functional differentiation among closely related strains. Limitations such as genome fragmentation, loss of strain-level resolution, incomplete recovery of complex genomic elements, and difficulty linking genes to specific taxa all point to a need for improved sequencing methods. These challenges motivate the use of new technologies that can deliver a more complete and high-resolution view of soil microbial communities.

1.4 Long-read metagenomics: enhanced genome recovery and strain resolution

Third-generation sequencing technologies, especially Oxford Nanopore (ONT) and Pacific Biosciences (PacBio), have improved metagenomics by producing long reads which can span thousands to tens of thousands of base pairs. Illumina short reads often struggle with repetitive or conserved regions. In contrast, long reads can cover full genes and repeat-rich areas in one stretch. This makes genome assemblies much completer and more contiguous. For example, long-read sequencing of biological soil crusts produced very long assemblies. It recovered almost 3,000 BGCs, including 712 that were full-length and complete (Van Goethem et al., 2021). These BGCs, which are often tens of kilobases long and produce secondary metabolites like antibiotics, are usually fragmented in short-read assemblies. Long reads captured full clusters on single contigs and revealed functional elements that short reads often miss. The technology has also become more accurate and productive. Early ONT reads had high error rates, around 5-15%. Recent developments like Q20+ chemistry have significantly improved read accuracy. Meanwhile, PacBio HiFi reads with about 99.9%

accuracy, now enable highly reliable assemblies from even complex microbial communities (Zhang et al., 2023).

Long-read metagenomics excels at recovering genomic features that are often fragmented or lost in short-read assemblies. One key example is the ribosomal RNA (rRNA) operon, which includes the 16S, 23S, and 5S rRNA genes. These operons span approximately 1,500 base pairs and contain repeated sequences that frequently cause breaks or collapses during short-read assembly. In contrast, long-read technologies can span entire rRNA operons in a single read, enabling complete recovery and accurate taxonomic assignment. For instance, in a study of the rice phyllosphere, Nanopore sequencing successfully assembled 669 complete 16S rRNA genes and linked them to 166 genomic species bins (Masuda et al., 2024). Notably, more than 70% of these bins (121) had less than 97% identity to any known 16S sequence, revealing novel microbial lineages that would have been undetectable with conventional methods. This highlights the power of long-read sequencing for high-resolution taxonomic profiling. By providing nearly complete rRNA operons directly linked to reconstructed genomes, it enables accurate placement of uncultured microbes within the microbial tree of life.

Second, plasmids and viruses are often missed or fragmented in short-read assemblies but can be more fully recovered using long-read sequencing. In the rice phyllosphere study, long-read data yielded 142 circular contigs, many of which were identified as novel plasmids and bacteriophages(Masuda et al., 2024). Similarly, Benoit et al. (2024) showed that combining PacBio HiFi reads with their metaMDBG assembler doubled the number of plasmids and viruses recovered compared to earlier short-read methods. This improvement is especially important because plasmids and phages often carry ecologically significant genes, including those involved in antibiotic resistance, specialized metabolism, and horizontal gene transfer. By capturing these mobile genetic elements more completely, long-read metagenomics enhances our ability to track microbial functions, understand community dynamics, and study how microbes adapt to changing environments.

Third, CRISPR arrays and other repetitive genomic regions can be fully resolved using long-

read sequencing. Long reads can span entire CRISPR loci, accurately reconstructing both the repetitive repeat sequences and the unique spacer regions that are often collapsed in short-read assemblies (Van Goethem et al., 2021). This advantage also extends to other repeat-rich elements, including transposons, pathogenicity islands, and additional BGCs beyond those initially mentioned. The complete recovery of BGCs represents a major advance, as it allows researchers to directly link these large gene clusters to their host genomes and even to gene expression profiles. This opens new opportunities for the discovery of novel natural products, such as antibiotics and other secondary metabolites. By capturing these structurally complex and ecologically important elements in full, long-read metagenomics greatly improves both functional annotation and the biological interpretation of environmental microbiomes.

Specialized assembly tools have been developed for long-read metagenomics. metaMDBG (for PacBio HiFi) and nanoMDBG (for Oxford Nanopore) address characteristic challenges in metagenomic data, including uneven coverage, strain variation, and sequencing errors (Benoit et al., 2024, 2025). These tools work better than older assemblers. For example, metaMDBG produced up to twice as many high-quality, circular genomes in complex samples. It also did well at recovering low-abundance genomes and small elements like plasmids and viruses. The assemblers aim to separate closely related strains rather than collapsing them into consensus sequences. This helps reveal more detail about microbial diversity at the strain level. nanoMDBG is built to handle the higher error rates of Oxford Nanopore reads. It stays efficient and can run large assemblies on modest computers. This makes it well-suited for large environmental studies. Some studies report that nanoMDBG performs as well as PacBio HiFi assemblies, while being more practical and cost-effective to use for large-scale soil microbiome work.

Third-generation sequencing has improved how we study metagenomes. It helps us build genome assemblies that reach or nearly reach full chromosomes. This allows us skip some older steps, like extensive binning, and focus directly on genomic differences. These technologies also allow analysis at the strain level. Researchers can now ask which strains, not just which species, carry certain genes and how those strains change across environments or treatments. This matters in soil systems, where microbial diversity is very high and many

lineages are hard to grow or study. With this enhanced resolution, we can connect functional genes, like those for enzymes in carbon cycling, to specific strains. We can also track how those strains respond to different soil practices or conditions.

1.5 Present study: Research gap and approach

Research Gap

Despite the advances in long-read metagenomics, a key research gap remains in applying these tools to soil ecosystems under different experimental treatments. To date, only a few studies have applied long-read assembly to soil or rhizosphere communities, e.g. a soil crust study (Van Goethem et al., 2021), or an agricultural soil inoculum trial (Buss et al., 2025). Not many studies so far have used long-read metagenomes across multiple soil samples to test how microbial genomes and their functions change with treatment (e.g. different fertilization regimes, plant rhizosphere vs bulk soil, etc.). Short-read studies have tried to explore these questions by looking at MAG abundance, but they often collapse strain differences and give only a rough view of how microbial communities respond. This limits our ability to connect specific microbial genomes and the functional genes they carry to soil carbon processes under different conditions. Furthermore, the living soil pathway framework emphasizes root-microbe interactions as central to carbon stabilization, yet most metagenomic studies lack the genomic resolution needed to link rhizosphere microbial functions to carbon outcomes at the strain level.

Study Aims and Objectives

This study addresses these gaps by applying Oxford Nanopore long-read metagenomic sequencing to reconstruct high-quality, near-complete microbial genomes from diverse soil metagenomes. The primary aim is to link microbial community structure (which organisms are present) and function (their metabolic pathways and ecological strategies) to environmental variables across multiple soils and experimental treatments. We focus on rhizosphere and bulk soil samples spanning three experiments: (1) Boorowa soil cores collected in 2019 and 2023, providing temporal comparisons; (2) plant culture trials with

different crop species, examining host plant effects; and (3) forest versus river soil barley rhizosphere samples, contrasting distinct soil types. By building a comprehensive, unified long-read genome catalogue across all experiments, the study creates a common reference framework for mapping, functional annotation, and cross-site differential analyses.

The study has two specific objectives. First, generate a unified long-read genome catalogue by co-assembling high-quality ONT reads from all three experiments, creating a comprehensive, non-redundant reference for consistent mapping and annotation across all samples. Second, quantify differential abundance of microbial genomes across environmental gradients (space, time, plant hosts, treatments) to identify which taxa and functions respond to specific conditions.

Chapter 2

De Novo Genome Assembly Using Long Read Sequencing

2.1 Methods

2.1.1 Sample collection and sequencing

We implemented three complementary experiments spanning sites, years, host plants, and treatments to examine microbial responses across environmental gradients.

(1) CSIRO Project - Boorowa soil temporal study (Alek Meade, 2019 & 2023). Soil cores were sampled along a 300-meter transect at 5-cm intervals from 0-20 cm depth, yielding 113 samples (55 in 2019 as baseline; 58 in 2023 after applications). For each sample, we measured electrical conductivity, pH, total nitrogen, total carbon, and moisture. DNA was extracted using the DNeasy PowerSoil kit, then tailed and barcoded using ligation sequencing. Libraries were sequenced on ONT R10 flow cells. Basecalling, trimming, and demultiplexing were performed with Dorado v1.1.0 in super-accurate mode (sup). This run produced 18.0 million reads totalling 39.16 Gbp of raw sequence with a mean read quality of Q14.2.

(2) Plant culture experiment (Viraj Kolhapuri, 2025). Two batches from the same soil examined plant-microbe interfaces and DNA workflow performance. Batch 1 comprised 35 samples across barley (cultivated and wild), canola, carinata, chickpea, and medicago, with shoot biomass, DNA concentration, and library preparation metrics recorded. Batch 2 comprised 35 samples across six host groups and controls (5 replicates each), adding soil EC, pH, bead type for extraction, and detailed library inputs. DNA was extracted using the DNeasy PowerSoil Pro Kit and barcoded with SQK-NBD114-96. Sequencing was carried out on PromethION with R10.4.1 chemistry. Basecalling, trimming, and demultiplexing used Dorado v1.1.0 (sup). Across both batches, we generated 28.3 million reads (75.55 Gbp, Batch 1) and 31.3 million reads (96.33 Gbp, Batch 2), with mean read qualities of Q13.3 and Q14.3, respectively.

(3) Forest (Kowen) vs river soil experiment (Saige Waugh, 2025). We collected 26 samples across biofertilizer, chemical fertilizer, and water controls, plus no-plant and biofertilizer-only references. We recorded DNA stock concentration, enzyme precipitation efficiency, and barcoded input volumes. DNA extraction used DNeasy PowerSoil Pro, and barcoding used SQK-NBD114-96. Sequencing employed PromethION (R10.4.1), with basecalling, trimming, and demultiplexing via Dorado v1.1.1 (sup). This dataset yielded 40.5 million reads totalling 55.67 Gbp at a mean read quality of Q13.0.

2.1.2 Read quality control

We performed quality control on raw FASTQ files to ensure only long, accurate sequences progressed to downstream analyses. First, we visualized read length, quality, and distribution with NanoPlot v1.44.1. We then assessed per-base quality metrics with FastQC v0.12.1, examining sequence quality profiles, GC and N content, length distributions, duplication levels, over-represented sequences, and k-mer content. Dorado trimming removed adapters, which we confirmed by screening over-represented sequences. Summaries from both tools were aggregated with MultiQC v1.31 to provide a single, comparable report across runs and batches. We also evaluated duplication rates and barcode cross-talk.

Filtering and trimming were carried out with Chopper v0.11.0 using two task-specific profiles. For external database alignment (e.g., to reference SMAG catalogues), we retained reads with minimum length 1,000 bp and minimum quality Q10 (≈90% accuracy). This profile preserves breadth while removing short, error-prone sequences that inflate false positives. For assembly, we applied stricter filters: minimum length 2,000 bp and minimum quality Q18 (≈98.4% accuracy). This enriches for long, high-fidelity reads that stabilize the assembly graph, reduce artifacts and improve consensus accuracy.

These filters were applied to all datasets. For example, Plant Culture Batch 2 produced an alignment-ready read set under the 1,000 bp/Q10 profile, while the assembly-ready pool (2,000 bp/Q18) yielded 123.85 Gbp with mean read quality Q21 and N50 = 5,111 bp. This two-tier strategy balances sensitivity for broad detection (alignment) with stringency for contiguity and correctness (assembly), improving downstream co-assembly, competitive

mapping, and functional annotation.

2.1.3 Contamination removal

To minimize host plant contamination while retaining microbial diversity, we applied a competitive alignment workflow to quality-filtered reads. All reads were mapped against a combined reference comprising representative plant genomes and a broad soil-microbe catalogue (SMAGs).

The plant panel included focal crop hosts: *Triticum aestivum* (bread wheat, GCA_018294505.1), *Hordeum vulgare* (barley, GCA_904849725.1), *Medicago truncatula* (medicago, GCA_003473485.2), *Cicer arietinum* (chickpea, GCA_000331145.2), *Brassica napus* (canola, GCA_020379485.1), and *Brassica carinata* (carinata, GCA_016771965.1), as well as *Eucalyptus regnans* (assembled by Scott Ferguson) for the forest control context.

Quality-filtered ONT reads were competitively aligned to this combined reference using minimap2 v2.30 (`-x map-ont`) and parsed with SAMtools v1.22 and pysam. For each read, we recorded the best plant hit and best SMAG hit using a tie-break score tuple (alignment score AS, mapping quality MAPQ, aligned length). Reads were then assigned to one of three classes: (i) plant-unique (best hit only to plant), (ii) microbe-unique (best hit only to SMAG), or (iii) ambiguous (valid hits to both with equal best score). Secondary and supplementary alignments were included by default to accommodate multi-segment ONT mappings. Ambiguous reads were retained for later probing against the custom long-read catalogue. Unmapped reads were also retained for downstream novelty discovery. This competitive approach enables downstream estimation of plant:microbe signal ratios (after organellar filtering), serving both as a quality assurance indicator and an ecological proxy for rhizosphere influence across sites, hosts, and treatments.

For each sample, we measured the proportion of reads mapping to plant and microbial references. This microbe-to-plant ratio served both as a quality control metric and as an ecological variable in downstream analyses.

2.1.4 Co-assembly strategy

To maximize genome recovery, we used a unified co-assembly approach where reads from samples within comparable experimental conditions were pooled before assembly (co-assembly approach). This method increases effective sequencing depth for low-abundance organisms, allowing recovery of genomes that would remain fragmented in single-sample assemblies. Reads from each individual sample were then mapped back to the unified assembly to estimate contig abundances for differential analysis.

We directly assembled long reads from complex communities to obtain high-contiguity genomes, potentially bypassing conventional binning. The rationale is that long-read assemblies can produce contigs orders of magnitude longer than short-read contigs, often spanning entire chromosomes or plasmids. These lengthy contigs inherently segregate different genomes, making binning (which clusters short fragments) less critical. In some cases, assembler output can be treated as final genomes. For example, a single circular contig ~4 Mbp in length with the expected complement of core genes is essentially a finished MAG. Long-read sequencing can thus "circumvent the requirement of generating MAGs in some cases, as large genome segments are captured directly... favoring low-abundance species" that short-read approaches would miss (Cuscó et al., 2021).

In practice, many pipelines still apply light binning or classification to group multiple contigs if a genome is split or to separate small plasmid contigs from chromosomes. However, the primary work of separating genomes is accomplished by the assembler leveraging read length.

2.1.5 Metagenomic assembly with nanoMDBG

We selected nanoMDBG for metagenomic assembly because it is specifically designed to handle Oxford Nanopore's error profile while maintaining computational efficiency for large, complex datasets (Benoit et al., 2025). nanoMDBG employs a minimizer-space de Bruijn graph (MDBG) approach to efficiently handle ONT's higher error rates while maintaining computational efficiency. The algorithm converts reads into ordered lists of representative k-mers (minimizers) and performs error correction entirely in minimizer space through binning and partial-order alignment. This enables processing of reads with up to ~4% error rates

without extensive pre-correction steps (Ekim et al., 2021). Corrected minimizer reads are then assembled through iterative construction of mini de Bruijn graphs with progressive abundance filtering. This filtering strategy simplifies the graph by removing low-abundance spurious nodes while preserving genuine genomic variation, facilitating separation of closely related strains. The final assembly graph is converted back to nucleotide space and polished. This design allows nanoMDBG to scale to terabase-scale co-assemblies on modest computational resources, typically completing within days on standard hardware, while producing assemblies with high contiguity, preserved operons, and effective strain-level resolution (Benoit et al., n.d.).

The merged, filtered reads (minimum length 2,000 bp and minimum quality Q18) were assembled using nanoMDBG v1.2 with the following parameters: `--in-ont --min-read-overlap 300 --min-abundance 1`. We tested several settings and selected these as a good balance of contiguity, read support, and runtime for the merged, multi-experiment dataset. The `--in-ont` flag activates Nanopore-specific handling. We set `--min-read-overlap 300` because our ONT reads had an N50 of approximately 5,111 bp with many medium-length reads in the distribution. A 300-bp minimum overlap retains more legitimate overlaps from these reads, improves contiguity in low-to-moderate coverage regions, and maintains connectivity across repetitive or error-prone zones. The `--min-abundance 1` parameter retains singleton tokens (k-min-mers seen once) to preserve low-abundance paths representing rare taxa, while still relying on nanoMDBG's minimizer-space correction and progressive filtering to suppress errors. This compromise kept more potentially informative rare signal than a stricter cutoff (e.g., 2) while maintaining manageable graph complexity and stable performance. All other parameters were left at defaults. After assembly, we polished the contigs using the `polish` command.

2.1.6 Post-assembly quality control

We summarized contig sizes and coverage, faceting results by circularity (circular versus non-circular contigs). We reported total assembly size and N50, then computed per-status N50 and the number of contigs ≥1 Mb per status. Results were visualized using a log-scaled length

histogram with N50 overlays. Coverage-length structure was profiled with hexbin plots, and coverage distributions were shown as violin plots by circularity. We enumerated and exported contig ID sets for thresholds used downstream: ≥10 kb OR ≥10× coverage, ≥20 kb OR ≥20× coverage, ≥1 Mb, and circular contigs (and their intersections) for downstream analysis.

We also ran MetaQUAST v5.3.0 in reference-free mode, using our assembly-ready reads to assess how well the assembly is supported by the data (Mikheenko et al., 2016). We evaluated only contigs ≥1 kb. MetaQUAST predicted genes, searched for rRNA sequences, and scanned for conserved single-copy markers to provide content-based quality checks. It flagged large structural problems and reported standard assembly statistics (N50, N90), total assembled bases, unsupported sequence, counts of mis-assemblies, and tallies of genes, rRNAs, and markers. These metrics represent a major complexity reduction from the raw input data while highlighting the substantial diversity remaining in rare microbial genomes.

2.2 Results

2.2.1 Assembly quality and contig characteristics

The nanoMDBG assembly produced 556,967 contigs ≥1 kb, totalling 7.69 Gbp with an N50 of 15.5 kb (Supplementary Table 1). Coverage depth averaged 16×, but the distribution was strongly skewed: over half the assembly (54.7%) fell below 10× depth, indicating a mixture of abundant core taxa and numerous rare community members (Supplementary Table 1).

The assembly yielded 86 contigs ≥1 Mb, with the longest reaching 4.52 Mb. We recovered 923 putatively complete circular replicons (Figure 2.1), of which 448 met our quality thresholds (≥10 kb or ≥10× coverage). For downstream analyses, we used contigs meeting these quality thresholds (n=268,290; 5.66 Gbp), which captured 74% of the total assembly.

Length distributions revealed distinct patterns between circular and non-circular contigs (Figure 2.1). Non-circular contigs displayed a unimodal distribution centred near the assembly N50, reflecting fragmentation across diverse low-abundance genomes. In contrast, circular contigs exhibited a bimodal distribution with two clear peaks: one at ~10 kb (potential plasmids and small replicons) and another at ~40 kb (small chromosomes or mega-plasmids). A sparse tail extended to 4.5 Mb, representing large bacterial chromosomes. Large contigs (≥1 Mb) included both non-circular chromosomes (n = 78) and circular replicons (n = 8), indicating recovery of fragmented genomes as well as complete genetic elements.

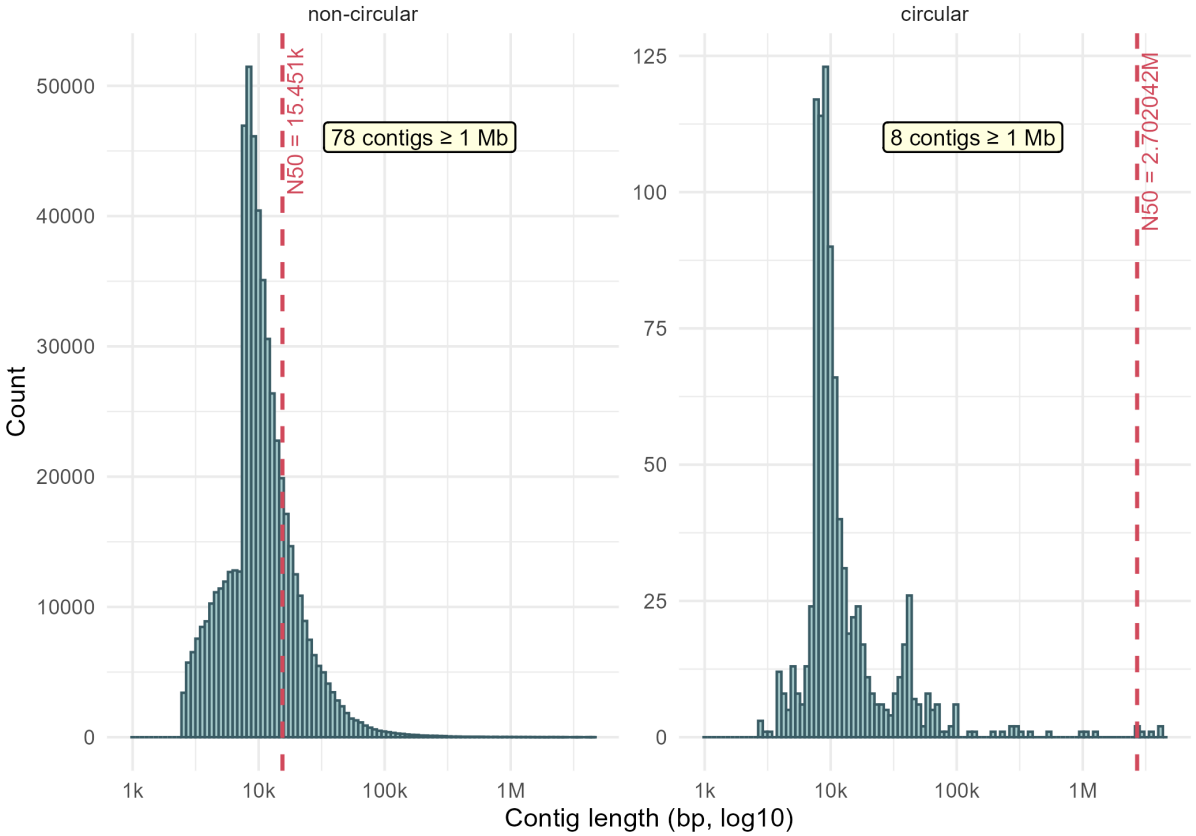


Figure 2.1 Contig length distribution by circularity status. Histograms show contig lengths ($\log_{10}$ scale) for non-circular (left panel, n=556,044) and circular (right panel, n=923) contigs. Red dashed lines indicate N50 values (non-circular: 15.5 kb; circular: 2.7 Mb). Box annotations show counts of contigs ≥1 Mb (78 non-circular, 8 circular). Non-circular contigs display a unimodal distribution centred near the assembly N50, whereas circular contigs show a bimodal pattern with peaks around ~10 kb (plasmids) and ~40 kb (small chromosomes), along with a sparse tail extending to 4.5 Mb that represents large bacterial chromosomes.

Coverage distributions were similar between circular and non-circular contigs, both spanning 1× to >100× with comparable medians around 3-5× (Figure 2.2). Circular contigs showed a subtle right skew with an extended upper tail.

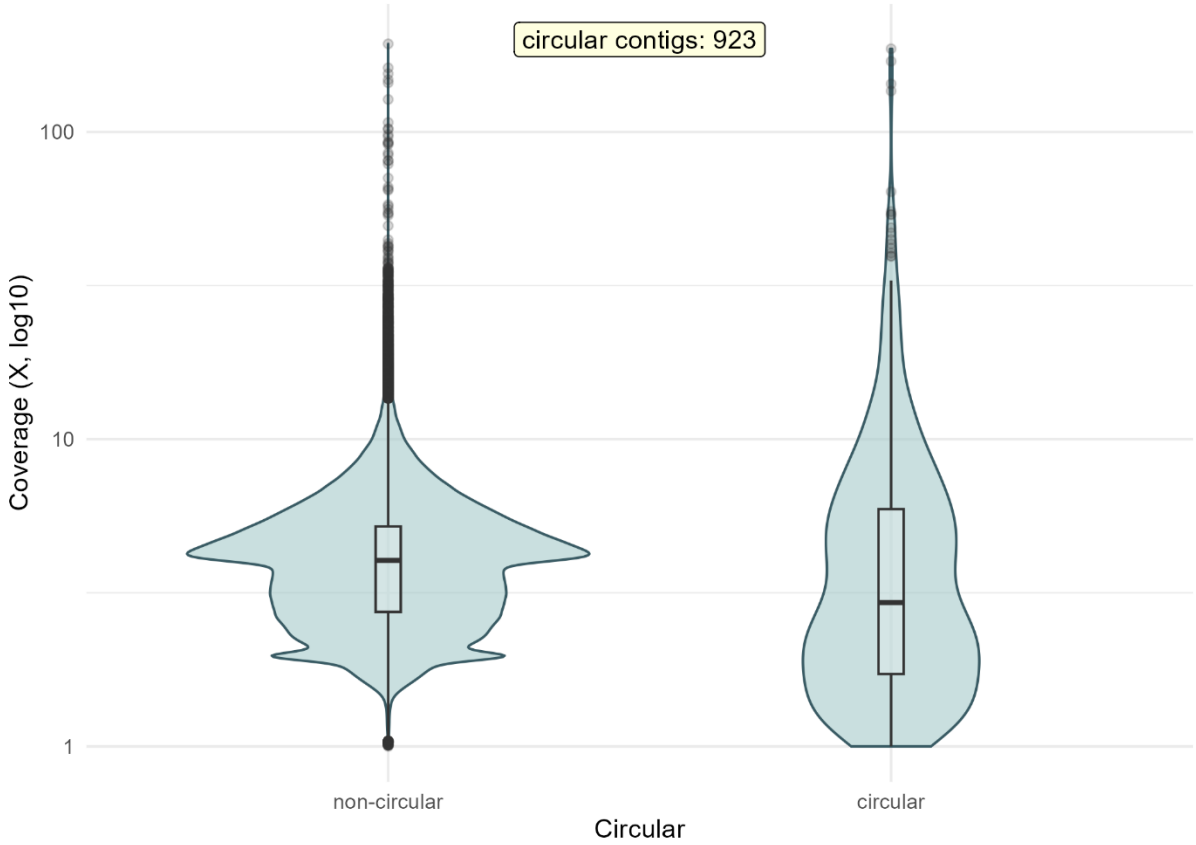


Figure 2.2 Coverage depth distribution by circularity status. Violin plots display read coverage depth (log10 scale) for non-circular (n=556,044) and circular (n=923) contigs. Box plots inside violins show median (horizontal line), interquartile range (IQR, box), and whiskers extending to 1.5× IQR. Grey dots indicate individual contigs.

The length-coverage hexbin plot revealed three distinct contig populations (Figure 2.3). The dense core (yellow-green hexagons) centred around 10-100 kb length and 3-10× coverage contained most sequences. This represents moderately well-assembled fragments from the diverse community. Sparse hexagons in the upper-right region represented high-quality sequences, which are long, well-covered contigs likely corresponding to abundant organisms with near-complete genome assemblies. The lower-left cluster of short, low-coverage contigs represented fragmented sequences from rare taxa. Our quality thresholds (10 kb length or 10× coverage; red dashed lines) effectively separated the reliable core assembly from the fragmented tail.

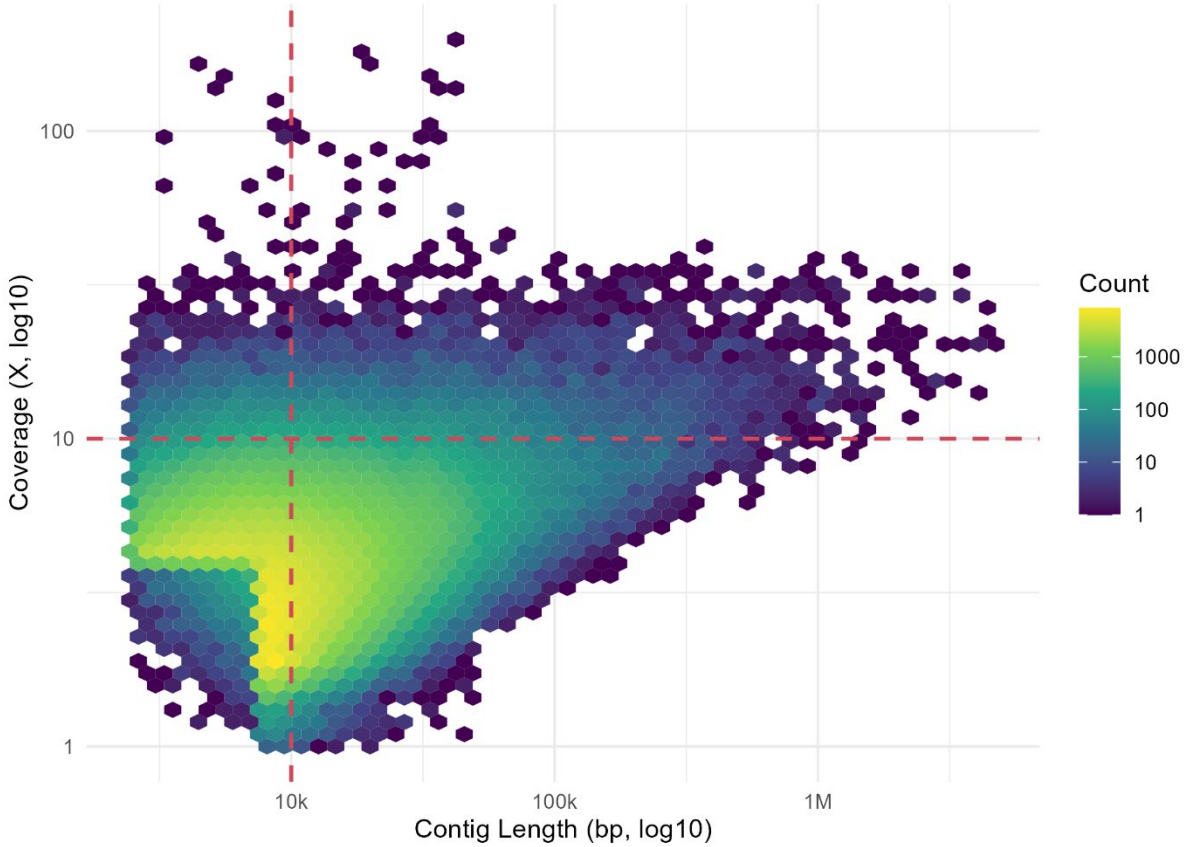


Figure 2.3 Relationship between contig length and coverage depth. Hexbin density plot shows the joint distribution of contig length and coverage depth (both log10 scale) across all 556,967 contigs. Color intensity represents contig count per hexagonal bin. Red dashed lines mark quality thresholds (10 kb length, 10× coverage).

Several metrics validated the assembly quality. Read remapping yielded a 237% alignment rate, indicating multi-mapping across conserved genomic regions shared among diverse community members. This is a typical pattern in complex metagenomes. The assembly contained 2,311 complete and 333 partial rRNA genes, suggesting the presence of hundreds to thousands of distinct organisms. Mean GC content was notably high at 66.0%. MetaQUAST evaluation confirmed strong read support, with 99.8% of assembled bases covered at ≥1× depth and no uncalled bases (Supplementary Table 1).

Together, these metrics indicate a high-quality assembly that captures both abundant core community members and rarer taxa, with sufficient coverage and contiguity for reliable taxonomic and functional annotation.

2.2.2 Comparison with the SMAG Catalogue

We compared our long-read assembly to the SMAG catalogue, which represents the current standard for soil microbiome genomics. The SMAG catalogue aggregated 3,304 globally distributed soil metagenomes into 21,077 species-level genome bins using Illumina short-read sequencing.

Our assembly N50 reached 15.5 kb, exceeding SMAG's 12.9 kb by 20% (Supplementary Figure 1A). The longest contig in our assembly measured 4.52 Mb, surpassing SMAG's maximum of 3.45 Mb by 31%. Most notably, we produced 86 contigs ≥1 Mb from 556,967 total contigs (0.015% rate), compared to SMAG's 177 from 9,280,391 total contigs (0.0019% rate), an 8-fold higher rate of large contigs despite having 16.7-fold fewer total sequences.

Length distributions differed between the two assemblies. SMAG's contig length distribution was strongly right-skewed with a peak at 2-3 kb (Supplementary Figure 1A), typical of short-read assemblies. In contrast, our assembly showed better representation of longer sequences (Figure 2.1). Analysis of SMAG's binned MAGs revealed that typical genomes comprised 100-500 contigs totalling 3-5 Mb (Supplementary Figure 2), indicating substantial fragmentation even after quality filtering for ≥50% completeness. The dense core of SMAG's distribution centred around 100-1,000 contigs per genome.

Our assembly produced 923 putatively complete circular replicons, while this feature is not reported for SMAG (Figure 2.1). These circular contigs exhibited a bimodal size distribution with peaks at 10 kb and 40 kb, plus a sparse tail extending to 4.52 Mb. Eight circular contigs exceeded 1 Mb.

We also achieved superior recovery of ribosomal RNA genes, with 2,311 complete and 333 partial rRNA sequences (Supplementary Table 1). In contrast, SMAG reported systematic absence of assembled rRNA genes in many otherwise near-complete MAGs.

These improvements: 20% higher N50, eightfold enrichment of megabase-scale contigs, complete rRNA recovery, and 923 circular replicons, show that long-read sequencing greatly improves genome completeness and annotation confidence compared with short-read

methods. The higher proportion of large contigs achieved with far fewer total sequences also indicates more efficient assembly, requiring less sequencing effort to reach superior results.

Chapter 3

Differential Abundance Analysis

3.1 Overview

Differential abundance (DA) analysis identifies microbial features (e.g. contigs, taxa or genes) whose abundances change in response to experimental conditions. In this chapter, we tested how shoot biomass affects microbial community composition using the Plant Culture Experiment, Batch 2 dataset, which spans six host groups (see section 2.1.1). Specifically, we asked: which contigs vary systematically with continuous shoot biomass?

This analysis can reveal microbes or functional genes linked to plant growth, illuminating how soil treatments affect the microbiome and ecosystem function. For instance, contigs that are more abundant in high-biomass soils may encode functions that support nutrient cycling and plant productivity. Larger plant growth should intensify rhizosphere processes, such as root exudation, microbial proliferation, and aggregate formation, which are central to the living soil pathway. In this way, DA analysis connects microbial diversity and function to the measured phenotype.

3.2 Methods

3.2.1 Count matrix construction

We constructed a contigs × samples count matrix by re-aligning Batch 2 reads to a filtered set of contigs. We retained contigs with length >10 kb or mean coverage >10× from the long-read assemblies. Clean reads (≥1,000 bp, Q ≥ 10) were mapped using minimap2 v2.30 with the ONT preset. We retained only primary alignments with MAPQ ≥ 20 (flag `-q 20`) and excluded unmapped, secondary, and supplementary alignments (flag `-F 0x904`) using SAMtools. From the filtered BAM files, we extracted a table of read-to-contig matches and aggregated read counts for each sample-contig pair.

To ensure adequate sequencing depth, we removed poor-quality samples identified in previous

quality control and further excluded libraries with fewer than 100,000 total mapped counts. This filtering left 26 samples for downstream analysis.

3.2.2 Data transformation and normalization

Metagenomic count data present three key challenges. First, the data are sparse, containing many zero counts. Second, they are over-dispersed, with variance exceeding the mean. Third, they are compositional, which means library size determines the count scale rather than absolute DNA amounts.

This compositional constraint creates statistical dependencies among features. When one contig increases in relative abundance, others must proportionally decrease, even if their absolute concentrations remain unchanged in the ecosystem. Standard statistical methods that assume independent observations can therefore produce misleading results.

To address these issues, we applied two complementary approaches, each suited to different analyses. For community-level analyses (ordination and PERMANOVA), we used compositional data analysis (CoDA) methods. For feature-level differential abundance testing, we used count-based modelling with edgeR.

Compositional data analysis for community structure

We applied the centred log-ratio (CLR) transformation to convert compositional data, which are constrained to a constant sum, into unconstrained values suitable for multivariate analyses in Euclidean space. The CLR transformation requires positive values, so we first replaced zeros using Bayesian-multiplicative replacement (zCompositions::cmultRepl in R) (Martín-Fernández et al., 2015; Palarea-Albaladejo & Martín-Fernández, 2015). This method estimates small positive values for zeros while preserving the relative structure of the data.

The CLR transformation inherently normalizes for library size by subtracting the log-geometric mean. Therefore, applying CLR to raw counts yields the same result as applying it to proportions. We did not perform rarefaction or total-sum scaling before CLR transformation.

We measured between-sample dissimilarity using the Aitchison distance, which is the

Euclidean distance calculated on CLR-transformed data. This distance metric provides a geometrically coherent measure of community differences that respects the compositional nature of the data.

Count-based normalization for differential abundance testing

For edgeR analysis, we kept the data as raw counts and normalized libraries using the trimmed mean of M-values (TMM) method, where M-values represent log-ratios of expression between samples. TMM estimates a scaling factor for each sample under the assumption that most contigs do not change across conditions. We specifically used TMMwsp (via `calcNormFactors(method = "TMMwsp")` in edgeR) because our data contain many zeros (Chen et al., 2025; Robinson et al., 2010). Unlike standard TMM, which discards contigs with zero counts in either sample when comparing libraries, TMMwsp pairs singleton counts across libraries, improving stability in sparse matrices. edgeR incorporates these scaling factors as offsets in the negative binomial generalized linear model.

3.2.3 Ordination and clustering

We performed principal component analysis (PCA) on the CLR-transformed count matrix to visualize sample patterns in an unsupervised manner (Abdi & Williams, 2010). PCA operates in Euclidean space, which matches the geometry of CLR-transformed data.

To quantify the effect of shoot biomass on community composition, we computed the Aitchison distance matrix and performed permutational multivariate analysis of variance (PERMANOVA) using vegan::adonis2 with the formula `distance ~ shoot_mass` (Anderson, 2017). This test reports R^2 (variance explained by shoot biomass) and a permutation-based p-value.

3.2.4 Differential abundance modelling with edgeR

We imported the count matrix and sample metadata (including continuous shoot_mass values) into edgeR in R. We first filtered low-prevalence contigs using filterByExpr to improve model stability, retaining only contigs present in a sufficient number of samples. We then applied TMMwsp normalization and estimated dispersions with estimateDisp.

We fit a generalized linear model using the quasi-likelihood (QL) F-test framework (glmQLFit) with shoot biomass as a continuous covariate (`design matrix: model.matrix(~ shoot_mass)`). We tested for association with shoot biomass using glmQLFTest on the shoot_mass coefficient. This analysis yields a $\log_2$ fold-change ($\log_2$FC) per unit increase in shoot biomass and a p-value for each contig (Chen et al., 2025).

3.2.5 Empirical false discovery rate calibration

Standard multiple testing correction using the Benjamini-Hochberg (BH) method assumes feature independence. However, compositional constraints and pervasive zero-inflation in metagenomic data violate this assumption, potentially leading to inaccurate false discovery rate (FDR) control (Yang & Chen, 2022).

We therefore applied a permutation-based calibration to derive empirical p-value thresholds (Li et al., 2022). Briefly, (i) we ran the edgeR-QL pipeline with the true shoot_mass values to obtain observed p-values; (ii) we performed 500 permutations, shuffling shoot_mass labels across samples and re-running the pipeline each time to generate a null distribution of p-values; (iii) for a grid of candidate thresholds τ, we calculated empirical FDR as:

$$\mathrm{FDR}_{emp}(\tau) = \frac{\text{mean number of discoveries } (\mathrm{p} \leq \tau) \text{ in permuted runs}}{\max(\text{number of discoveries } (\mathrm{p} \leq \tau) \text{ in real data}, 1)}$$

We selected the threshold τ* where $\mathrm{FDR}_{\mathrm{emp}}(\tau^*) \approx 0.05$ and reported contigs with $p \leq \tau^*$ as significant at an empirical 5% FDR (Xie et al., 2005) (Supplementary Figure 3 & Supplementary Figure 4). This approach accounts for correlation structure and non-standard null behaviour in the data, providing more reliable FDR control than asymptotic p-values alone. In practice, we found τ* to be slightly more stringent than the nominal BH-FDR < 0.05 cutoff, reflecting the additional conservatism needed under compositional and zero-inflated data.

3.2.6 Effect size thresholds and biological interpretation

To ensure biological relevance, we considered both statistical significance and effect size. Contigs were retained if $|\log_2\mathrm{FC}| \geq 0.5$, equivalent to a 1.4-fold change in abundance per unit

increase or a 0.71-fold decrease in shoot biomass. This threshold filters out trivial differences that may appear significant only because of high sequencing depth but lack ecological importance. The dual criteria of $p < \tau^*$ and $|\log_2 FC| \geq 0.5$ therefore identify contigs showing both statistically robust and biologically meaningful responses to plant productivity.

We calculated the expected fold change per unit increase in shoot biomass as:

$$FC_{\Delta x=1} = 2^{\log 2FC}$$

And the corresponding percent change as:

$$\%\Delta = (2^{\log 2FC} - 1) \times 100\%$$

3.2.7 Annotation of differentially abundant features

We aligned significant contigs to a combined reference of SMAG genomes and plant genomes using minimap2 (assembly-to-assembly preset, `-x asm5`) to assign taxonomic annotations. We retained primary alignments with MAPQ ≥ 20. We summarized taxonomic composition separately for contigs enriched in high-biomass versus low-biomass samples to identify clade-level trends.

3.3 Results

3.3.1 Community structure shows a biomass gradient but most variation is unexplained

Principal component analysis revealed that shoot biomass shapes community composition along PC1, though it represents only a minor portion of total community variation. PC1 and PC2 together captured just 11.4% of total variance (Figure 3.1).

The ordination showed that most high-biomass communities clustered on the negative side of PC1, creating a clear biomass gradient. The top marginal panel demonstrated a strong negative correlation between PC1 scores and shoot biomass ($R^2 = 0.485$, $p < 0.001$). This means that nearly half the variance along PC1 can be explained by plant growth. Samples with high shoot mass displayed low PC1 values, while samples with low shoot mass displayed high PC1 values.

In contrast, PC2 showed no relationship with shoot biomass ($R^2 = 0.075$, $p = 0.18$). Samples with varying biomass levels were distributed evenly across this axis, indicating that PC2 captures other sources of variation rather than plant productivity.

Host plant species did not drive community clustering. Samples did not segregate by host plant identity (represented by point shapes) (Figure 3.1).

We quantified the overall biomass effect using PERMANOVA on Aitchison distances. Shoot biomass explained only 4% of total community variance, and this relationship was not statistically significant ($p = 0.217$). This result aligns with the PCA findings when we consider the scales involved. PC1 explains 6.2% of total variance, and biomass explains 48.5% of PC1. Multiplying these values yields approximately 3% of total variance, which matches the PERMANOVA R^2 estimate. In summary, while the biomass signal along PC1 is real and substantial, it represents a small fraction of overall community variation. More than 95% of the variance arises from factors other than plant growth.

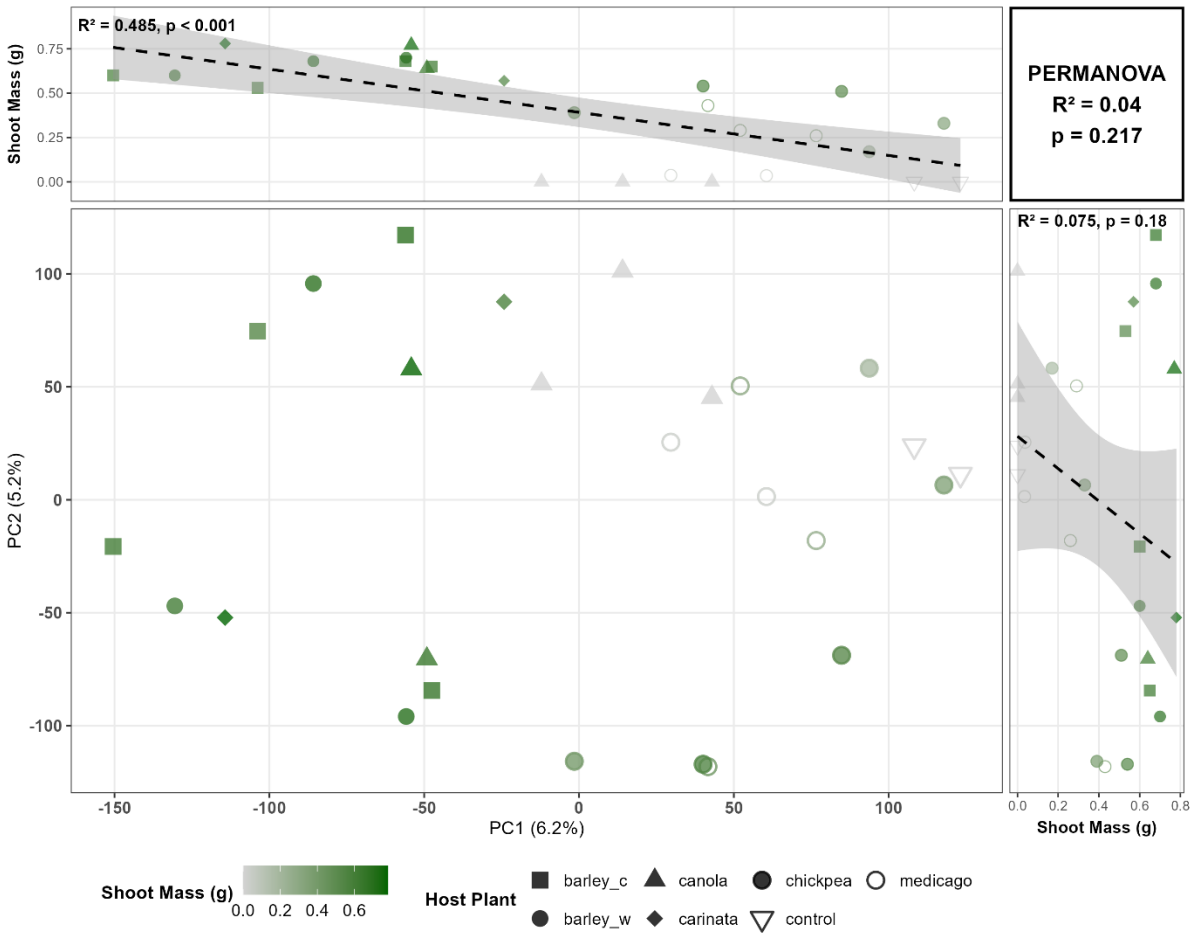


Figure 3.1 Principal component analysis of soil microbial communities based on CLR-transformed contig abundances (Aitchison distance). The main panel shows sample ordination on PC1 (6.2% variance explained) and PC2 (5.2% variance explained), with point color indicating shoot biomass and shape indicating host plant species. Top and right marginal panels show relationships between PC scores and shoot mass, with linear regression lines, 95% confidence intervals, R^2 values, and p-values. PERMANOVA results (top-right) test the overall effect of shoot mass on multivariate community composition ($R^2 = 0.04$, $p = 0.217$).

3.3.2 Differential abundance analysis identifies 194 biomass-associated contigs

Despite the small overall community effect detected by PERMANOVA, differential abundance analysis successfully identified individual contigs that varied systematically with shoot biomass. Of 3,358 contigs analysed, 194 showed significant associations: 108 were enriched in high-biomass samples and 86 were enriched in low-biomass samples (Figure 3.2).

All statistically significant contigs also exceeded the effect size threshold ($|\log_2 FC| > 0.5$), confirming that these differences are biologically meaningful. The volcano plot revealed that

significant contigs spanned a wide range of fold changes, with some showing more than 4-fold differences in relative abundance per unit increase in shoot biomass.

Many significant contigs were large, with lengths exceeding 350 kb and some surpassing 2 Mb. These lengths suggest high-quality metagenome-assembled genomes (MAGs) representing abundant and functionally important taxa. Interestingly, no circular contigs showed significant biomass associations.

We taxonomically annotated the top 20 significant contigs through alignment to the soil MAG database (Supplementary Table 2). Five contigs remained unmapped, potentially representing novel or poorly characterized lineages.

High-biomass-associated taxa included Actinobacteriota (genera *Pseudonocardia*, *Kribbella*, and family Micromonosporaceae), Chloroflexota (class Ellin6529), and multiple *Massilia* strains (Gammaproteobacteria). Four distinct *Massilia* contigs (0.78-2.4 Mb) were enriched at higher biomass.

Low-biomass-associated taxa included Alphaproteobacteria (families *Xanthobacteraceae* and *Reyranellaceae*), Myxococcota, Gemmatimonadota, and Acidobacteriota (class Blastocatellia, family *Pyrinomonadaceae*). Two *Pyrinomonadaceae* strains (2.3-2.8 Mb genomes) dominated low-biomass samples.

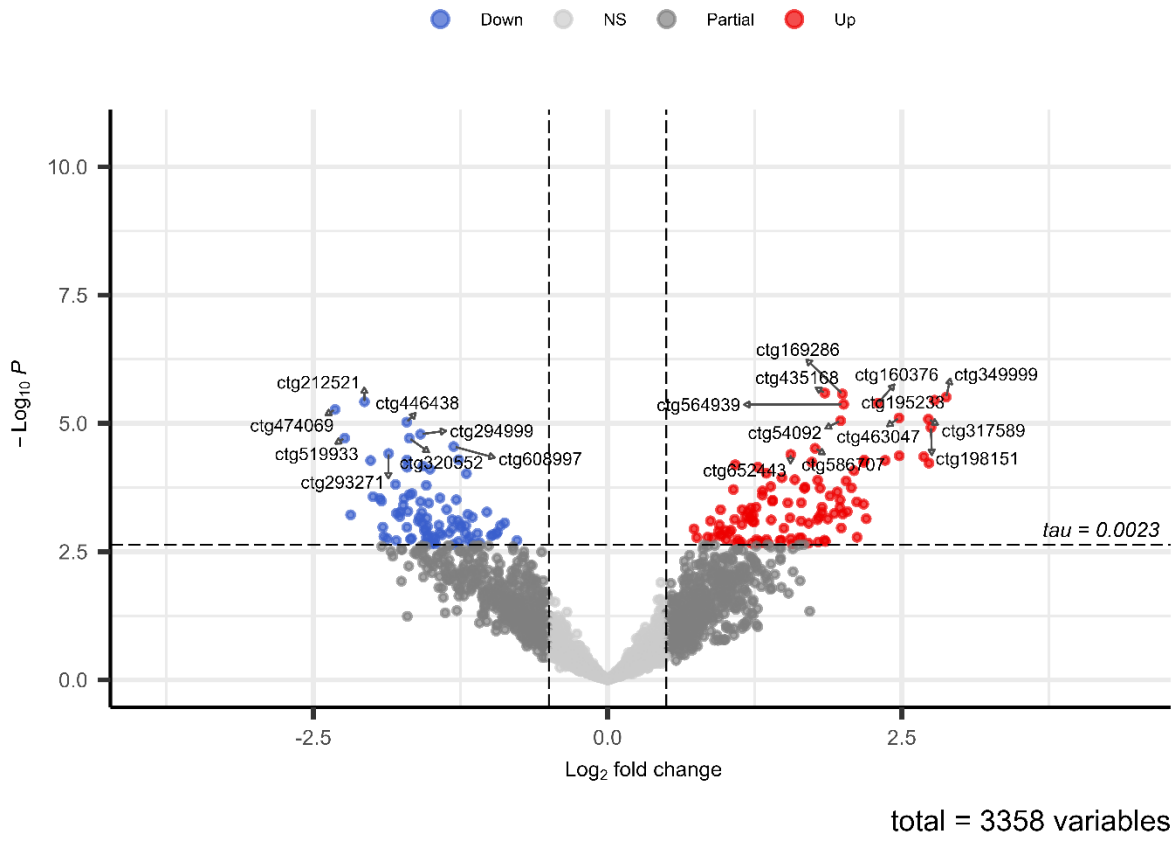


Figure 3.2 Volcano plot of differentially abundant contigs associated with plant shoot biomass. Differential abundance analysis of 3,358 metagenomic contigs (⩾10 kb or ⩾10x) using edgeR with empirical FDR control ($\tau = 0.0023$, equivalent to FDR < 0.05). Each point represents a contig, with x-axis showing $\log_2$ fold change and y-axis showing $-\log_{10}$ p-value. Contigs are coloured by significance: red (Up) = enriched in high-biomass samples ($\log_2FC > 0.5$, $p < \tau$); blue (Down) = depleted in high-biomass samples ($\log_2FC < -0.5$, $p < \tau$); dark grey (Partial) = significant for either fold change or p-value alone; light grey (NS) = not significant. Vertical dashed lines indicate $\log_2FC = \pm 0.5$ thresholds; horizontal dashed line indicates empirical p-value cutoff (τ). Top 20 significant contigs are labelled.

The heatmap confirmed these taxonomic patterns through sample clustering (Figure 3.3). Samples segregated naturally by shoot biomass, with the dendrogram closely corresponding to the biomass gradient. Low-biomass samples (left cluster) displayed high abundance of Gemmatimonadota, Acidobacteriota, and Alphaproteobacteria, while high-biomass samples (right cluster) were dominated by *Massilia* strains. This clustering validates the biological relevance of the statistical findings. One contig (ctg212521) could not be taxonomically assigned, potentially representing a novel lineage preferentially enriched under low-biomass conditions (Supplementary Table 3).

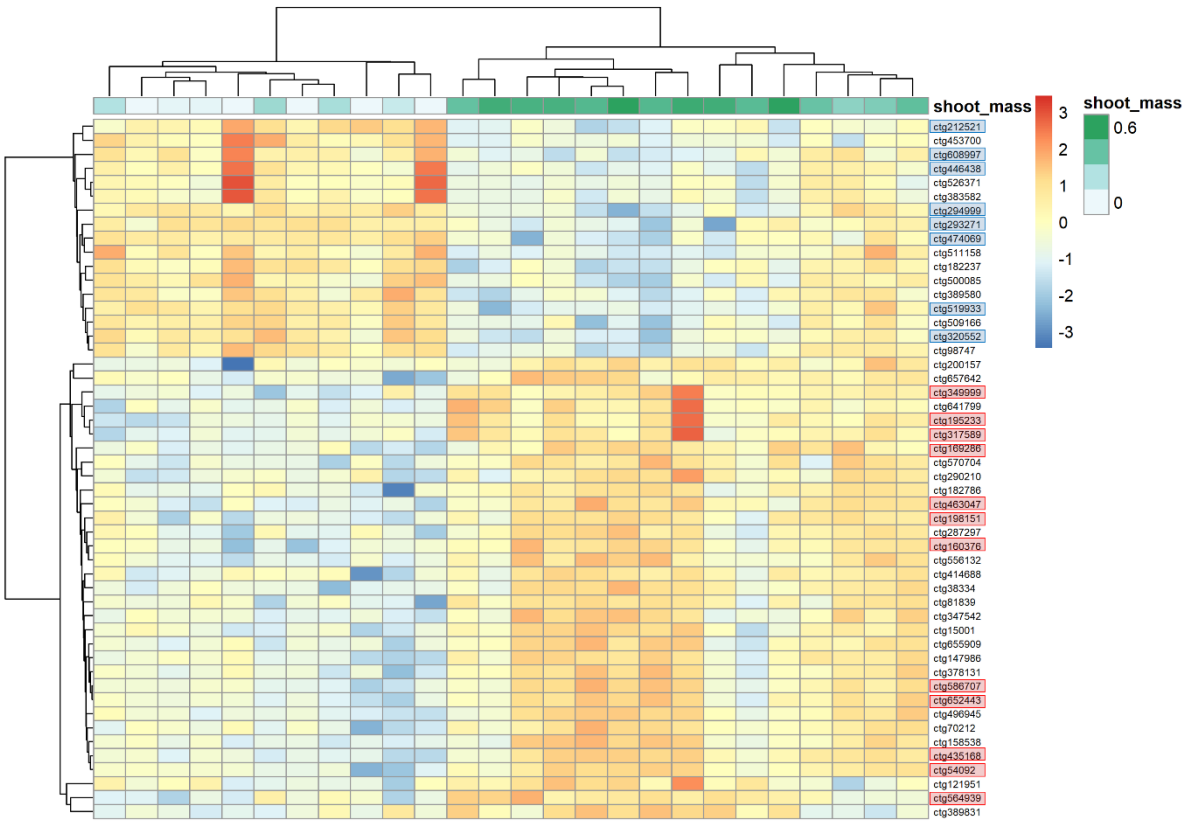


Figure 3.3 Hierarchical clustering heatmap of the top 50 differentially abundant contigs across plant shoot biomass gradient. Heatmap shows normalized abundance (TMM-normalized log-CPM values that were z-scored by contig across samples) of the 50 most significant contigs from differential abundance analysis. Columns represent individual soil samples (n = 26), clustered by 1 − Spearman correlation (average linkage), with shoot biomass annotated at top (blue = low biomass, red = high biomass). Rows represent contigs, clustered by abundance patterns. Contigs highlighted in red boxes indicate the top 20 most significant differentially abundant contigs ($p < \tau = 0.0023$, $|\log_2 FC| > 0.5$) from volcano plot analysis (Figure 3.2). Dark red cells indicate high relative abundance; dark blue indicates low abundance.

3.3.3 Large contigs show biomass associations across diverse coverage depths

We identified 11 differentially abundant contigs exceeding 1 Mb in length, representing high-quality assemblies of ecologically important taxa (Figure 3.4). These contigs ranged from 1.0 to 3.3 Mb and included both high-biomass-associated taxa and low-biomass-associated taxa (Supplementary Table 4). High-biomass-associated contigs showed consistently high sequencing coverage (20-34×), indicating abundant populations with near-complete genome assemblies (Supplementary Table 4). This group included three *Massilia* strains (Burkholderiales), two *Geodermatophilus* strains (Mycobacteriales), and one *LX47W* strain

(Burkholderiales). Low-biomass-associated contigs comprised two *Pyrinomonadaceae* strains (Acidobacteriota), two *SCGC-AG-212-J23* strains (Burkholderiales), and one taxonomically unassigned contig (1.0 Mb, ctg462036).

The large genome sizes (1.0-3.3 Mb) represent near-complete bacterial genomes. These 11 contigs span a wide range of coverage depths (5-34×) and include both high- and low-biomass-associated taxa. Differential abundance patterns were observed across various contig lengths and coverage levels, indicating that these effects were not limited to the largest or most deeply sequenced assemblies. Given their size, completeness, and ecological importance, these 11 contigs are strong candidates for detailed functional annotation.

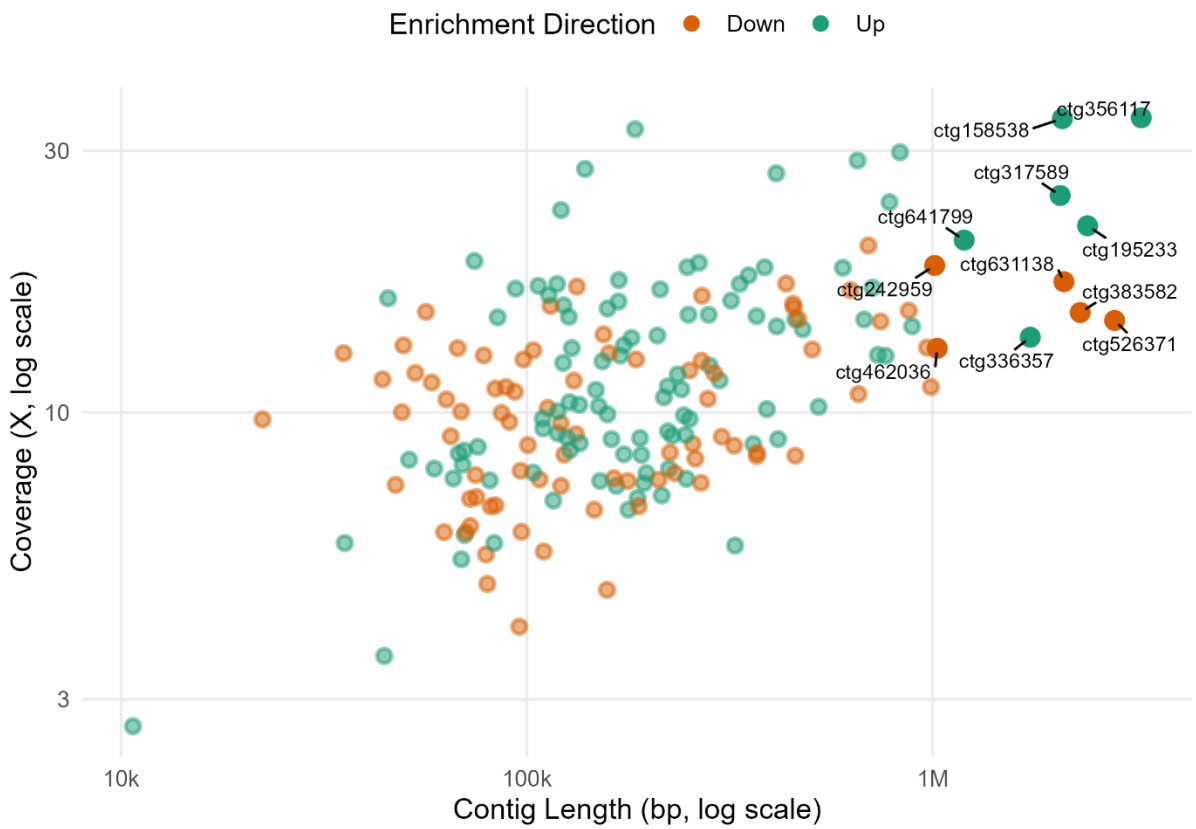


Figure 3.4 Contig size and coverage of long differentially abundant contigs (>1 Mb). Scatter plot shows the relationship between contig length (x-axis, log scale) and sequencing coverage (y-axis, log scale) for all differentially abundant contigs exceeding 1 Mb in length (n = 11). Points are coloured by enrichment direction: green (Up) = enriched at higher shoot biomass; orange (Down) = enriched at lower biomass. Top significant contigs are labelled.

3.3.4 Actinobacteriota dominate high-biomass communities

To comprehensively characterize the bacteria enriched at higher plant biomass, we taxonomically annotated all 108 differentially abundant contigs. Of these, 74 (68%) were successfully mapped to soil MAG databases (Figure 3.5).

Actinobacteriota dominated the high-biomass community, comprising 41 of 74 mapped contigs (55%). Within this phylum, the order Mycobacteriales represented 34 contigs (83% of Actinobacteriota), distributed among four main genera: *Pseudonocardia* (10 contigs), *Geodermatophilus* (6), *Actinoplanes* (6), and *Blastococcus* (5).

Proteobacteria represented the second major phylum (17 contigs, 23%), featuring diverse orders: Burkholderiales included *Massilia* (4 contigs) and *Nitrosospira* (4), while Azospirillales and Rhizobiales contributed *Skermanella* (3) and *Bradyrhizobium* (2), respectively.

Chloroflexota comprised 9 contigs, all from class Ellin6529 (order QHBO01) but unclassified at genus level. Minor phyla included Gemmatimonadota (2 contigs), Acidobacteriota (1), Myxococcota (1), Planctomycetota (1), and Patescibacteria (1).

A substantial fraction of differentially abundant contigs, 34 of 108 (32%), could not be assigned to reference genomes, highlighting the presence of novel or underrepresented lineages in these soil communities. Among the 74 successfully mapped contigs, an additional 15 (20%) remained unclassified below family level, further emphasizing the taxonomic novelty within these assemblies (Figure 3.2, Figure 3.3).

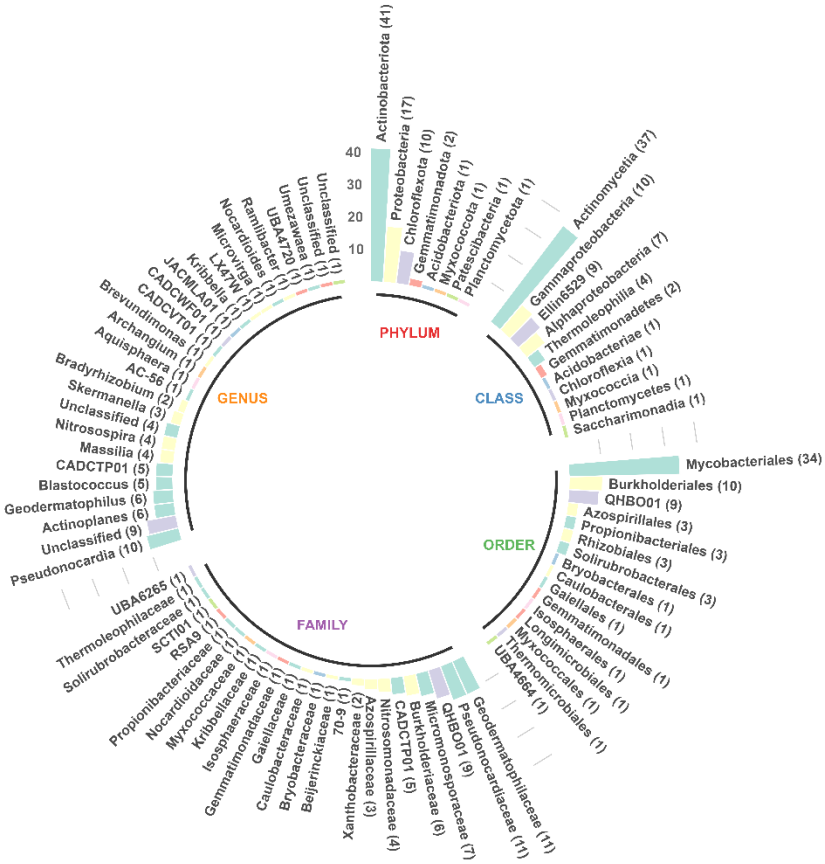


Figure 3.5 Taxonomic composition of microbial contigs enriched at higher plant biomass. Circular bar plot shows the hierarchical taxonomic classification of 74 mapped contigs (out of 108 total, ~68%) that were significantly enriched in high-biomass samples ($p < \tau = 0.0023$, $\log_2 FC > 0.5$). The plot displays counts at five taxonomic levels from phylum to genus, with the number of contigs for each taxon shown in parentheses. Bar colours distinguish different taxa at phylum level.

3.3.5 Proteobacteria dominate low-biomass bacterial communities with distinct oligotrophic lineages

To characterize bacteria associated with lower plant productivity, we taxonomically annotated all 86 contigs enriched at low biomass. Of these, only 36 (42%) were successfully mapped to soil MAG databases, compared to 68% mapping success for high-biomass contigs. This lower annotation rate suggests that low-biomass communities harbour more taxonomically novel or poorly characterized lineages (Figure 3.6).

Proteobacteria dominated the low-biomass community (16 contigs, 44%), but with a fundamentally different composition than high-biomass samples. Within Burkholderiales (9

contigs), the family SG8-39 (genus *SCGC-AG-212-J23*) accounted for 7 contigs, representing a distinct oligotrophic lineage. Alphaproteobacteria included *BOG-931* (3 contigs, Xanthobacteraceae), *Sphingomicrobium* (2), and *Reyranella* (1).

Gemmatimonadota represented the second major phylum (6 contigs, 17%), including *JACDDX01* (2) and *UBA4720* (2), confirming this group's role as an oligotrophic indicator. *Acidobacteriota* (5 contigs, 14%) comprised two *Pyrinomonadaceae* strains (PSRF01) and three *DSPA01* (Acidobacteriales), both characteristic slow-growing oligotrophs. Myxococcota contributed 5 contigs (14%), primarily *JABFXX01* (4), representing predatory bacteria adapted to nutrient-limited conditions.

Additional diversity included one archaeal contig (*Nitrosocosmicus*, an ammonia-oxidizing archaeon), *Firmicutes* (*Priestia megaterium*, a spore-forming bacterium), Verrucomicrobiota, and Chloroflexota. This taxonomic composition contrasts sharply with the Actinobacteriota-dominated high-biomass community (Figure 3.5), revealing fundamentally different community structures along the plant productivity gradient.

Figure 3.6 Taxonomic composition of bacterial contigs enriched at lower plant biomass. Circular bar plot shows the hierarchical taxonomic classification of 36 mapped contigs (out of 86 total, 42%) that were significantly enriched in low-biomass samples ($p < \tau = 0.0023$, $\log_2 FC < -0.5$). The plot displays counts at five taxonomic levels from phylum to genus, with the number of contigs for each taxon. Bar colours distinguish different taxa at phylum level.

3.3.6 Circular genetic elements show no biomass-associated patterns

To characterize circular genetic elements, including plasmids, small chromosomes, and organellar genomes, we identified 448 circular contigs (≥10 kb or ≥10× coverage) in the metagenome assembly. Of these, 211 (47%) were successfully mapped to reference databases (Figure 3.7).

Notably, none of these circular contigs showed significant differential abundance with respect to plant biomass ($p > \tau$), contrasting with the strong biomass associations observed for linear chromosomal contigs.

Eukaryotic sequences comprised 49 contigs (23% of mapped contigs), primarily representing plant chloroplast and mitochondrial genomes from crop species present in the soil

environment. These include *Hordeum* (barley, 16 contigs), *Brassica* (canola and carinata, 7), plus individual representatives from *Eucalyptus*, *Medicago*, and *Cicer* (chickpea).

Bacterial circular contigs (162 contigs, 77%) showed diverse phylum representation, with Proteobacteria (49 contigs, 23%), Actinobacteriota (31, 15%), *Firmicutes* (28, 13%), Chloroflexota (19, 9%), *Cyanobacteria* (12, 6%), and Gemmatimonadota (10, 5%) as major groups. Archaeal sequences were represented by eight Nitrososphaeria contigs (ammonia-oxidizing archaea).

The taxonomic composition of circular elements differed markedly from linear contigs. Firmicutes were substantially more represented among circular elements (13% versus rare in differential abundance analysis), and Cyanobacteria appeared uniquely abundant despite being absent from biomass-associated linear contigs. These compositional differences, combined with the absence of biomass-related abundance patterns, suggest that plasmids and other circular replicons respond to different selective pressures than chromosomal genomes.

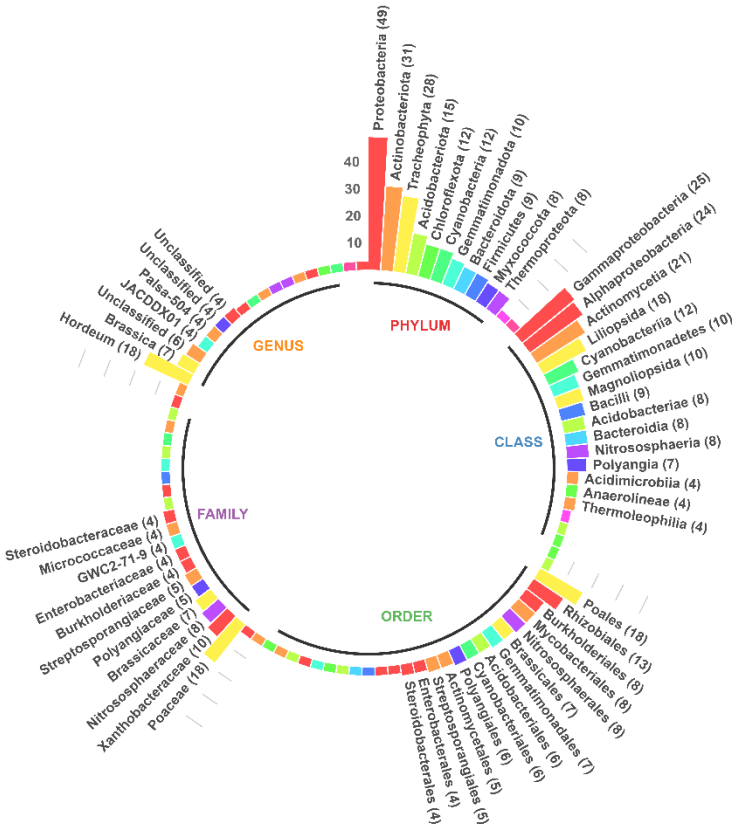


Figure 3.7 Taxonomic composition of circular genetic elements in soil metagenomes. Circular bar plot shows

the hierarchical taxonomic classification of 211 mapped circular contigs (out of 448 total, 47%) that were assembled from soil metagenomic data (≥10 kb or ≥10× coverage). The plot displays counts at five taxonomic levels from domain to genus, with the number of contigs for each taxon shown in parentheses. Bar colours distinguish different taxa at phylum level. Notably, no circular contigs were significantly differentially abundant with respect to plant biomass ($p > \tau$).

Chapter 4

Discussion

4.1 High-quality genome assembly provides foundation for functional inference

4.1.1 Assembly Quality Exceeds Short-Read Standards

Our Oxford Nanopore co-assembly recovered 86 contigs exceeding 1 Mb, with the longest reaching 4.52 Mb, approaching complete bacterial chromosomes. These results demonstrate that the technical advantages of long-read sequencing translate into tangible improvements when applied to complex soil systems. The assembly reflects soil's extreme diversity: over half of the assembled bases came from low-coverage taxa (<10× depth), representing rare community members. However, dominant organisms assembled into contigs approaching complete genomes, with 86 contigs exceeding 1 Mb. This achievement is particularly significant given that soil harbours thousands of microbial species at vastly different abundances.

Benchmarking against the global SMAG catalogue (Ma et al., 2023, see section 2.2.2) reveals substantial methodological gains. Our assembly achieved 20% higher N50 and 8-fold higher rate of recovering megabase-scale contigs, which achieved with far fewer total sequences. This indicates that long-read technology enables more complete genome reconstruction without requiring massive sampling efforts (Supplementary Figure 2). More importantly, we recovered 923 circular replicons representing finished genetic elements, a capability systematically absents from short-read assemblies. These closed sequences bypass the ambiguities inherent in fragmentary assemblies, providing unequivocal evidence of genome completeness.

These results validate predictions from prior long-read metagenomic studies in less complex systems (Cuscó et al., 2021; Masuda et al., 2024). The successful extension to soil suggests that genome-scale assembly from environmental samples is now technically feasible even for

extremely complex communities. This represents a methodological inflection point: we can now aim for complete genome recovery rather than accepting fragmented assemblies as inevitable.

4.1.2 Complete genome features enable confident functional inference

Recovery of 2,311 complete rRNA operons addresses a critical limitation that has plagued environmental genomics. Short-read assemblies systematically fragment these taxonomically essential markers, forcing reliance on indirect inference through less reliable genes (Ma et al., 2023). By embedding full-length rRNA genes directly within genomic contexts, our assemblies enable high-confidence taxonomic placement and cross-validation with amplicon-based research.

Beyond rRNA genes, preservation of gene synteny improves functional annotation accuracy. Complete operons, biosynthetic gene clusters, and mobile genetic elements provide genomic context that database searches alone cannot capture (Cuscó et al., 2021). For soil carbon cycling studies, this allows identification of full metabolic pathways—such as lignocellulose breakdown or exopolysaccharide formation—and linking them to specific organisms. This shifts analysis from detecting single genes to describing complete functional modules.

Additionally, rRNA copy number variation provides ecological insights. Organisms with many operons are typically fast-growing copiotrophs, while those with single copies tend to be oligotrophs (Liu et al., 2024). This distinction helps predict microbial life strategies that fit within the living soil pathway framework.

4.1.3 Co-assembly strategy balances sensitivity and strain resolution

Our co-assembly strategy combined reads from similar experimental conditions to increase sequencing depth and capture low-abundance organisms. This method helped assemble genomes that would otherwise remain fragmented in single samples. Many contigs in the final assembly showed coverage just above the minimum threshold (3-5×), suggesting they represent rare taxa that gained enough reads only after pooling. This kind of pseudo-enrichment has been effective in other complex microbiomes (Erkorkmaz et al., 2025) and

appears equally useful for soil communities.

However, co-assembly improves genome recovery but can reduce fine-scale strain resolution. Closely related strains may merge into shared consensus sequences, hiding small but important functional differences. The high read mapping rate we observed (237% relative to assembly length) suggests strong sequence similarity among related genomes. It is uncertain whether this pattern reflects real strain diversity or conserved regions shared between species. This limitation is important to note. We recovered population-level genomes for the main community members, but strain-level variation within these groups likely remains unresolved.

Future studies can use hybrid strategies. The initial co-assembly can capture the full range of genomic content, and later strain-resolved methods, such as long-read variant calling or assembly graph analysis, can separate population structure within genomes (Wick & Holt, 2022). This two-step approach keeps the sensitivity benefits of co-assembly while adding the strain-level detail that is now seen as ecologically important.

4.1.4 Genome-resolved foundation enables ecological applications

When benchmarked against global soil genome catalogues, our assembly shows clear improvements in contiguity, completeness, and recovery of difficult genomic regions. These advances help overcome technical barriers that have limited soil microbiome research for decades (sections 1.3-1.4). The value goes beyond better assembly numbers, since high-quality genomes make it possible to perform the functional analyses at the core of this study.

Complete genomes with preserved gene synteny allow confident linking of organisms to metabolic capabilities. For the living soil pathway framework (section 1.2), this means we can now identify which Actinobacteria or Proteobacteria strains carry genes for degrading specific root exudates, which organisms produce extracellular polymers that promote aggregation, or which taxa harbour biosynthetic clusters for compounds influencing plant-microbe signalling. Mapping reads from individual rhizosphere versus bulk soil samples back to this reference quantifies how these organisms shift along environmental gradients, transforming correlative descriptions into mechanistic understanding.

4.1.5 Key limitations and considerations

While these genomic advances offer substantial advances for ecological interpretation, several important limitations remain. First, many rare taxa remain only partly assembled, with more than half of the total bases coming from contigs under 10× coverage. Deeper sequencing could improve recovery, but the gain would likely be small because of the extreme diversity in soil. Although we achieved good contiguity (86 contigs ≥1 Mb), most organisms still assembled into several fragments rather than complete genomes. Using hybrid methods that combine long and short reads for polishing may help improve both completeness and accuracy (Wick & Holt, 2022).

A further limitation is that functional annotation depends on reference databases that are still incomplete for many environmental organisms. Since about 78% of soil species lack cultured representatives (Ma et al., 2023), many genes remain hypothetical or without clear functions. Expanding cultured genome collections and improving tools for function prediction are important next steps that go beyond the scope of this study.

Most importantly, our genome-resolved approach shows the metabolic potential of organisms, but not which ones are active. Metagenomics alone cannot tell whether populations are dormant or active, or which pathways respond to changes in the environment. Combining it with other methods such as metatranscriptomics, metaproteomics, and stable isotope probing can help fill this gap. For example, ^{13}C-labeling experiments that trace carbon from root exudates into specific microbial groups could directly test predictions about living soil pathway contributions based on genomic data (Sokol et al., 2022).

4.2 Plant-microbe interactions structure communities along productivity gradients

4.2.1 Plant productivity structures communities along a copiotrophic-oligotrophic gradient

Differential abundance analysis identified 194 contigs whose abundances changed

consistently with plant shoot biomass. The results showed a clear pattern: shoot biomass explained only 4% of the total community variation (not significant by PERMANOVA) but accounted for 48.5% of the primary ordination axis (PC1) (Figure 3.1). This suggests that plant productivity enriches certain microbial groups rather than reshaping the whole community.

High-biomass soils were dominated by Actinobacteriota (55% of mapped contigs), including fast-growing genera such as *Pseudonocardia* and *Massilia*. Low-biomass soils contained more oligotrophic groups like *Acidobacteriota* and *Gemmatimonadota* (Supplementary Table 2). These taxonomic shifts match established ecological theory on resource availability and microbial life strategies, showing how plant productivity helps shape rhizosphere microbial communities.

4.2.2 Rhizosphere processes drive microbial community turnover

The strong link between PC1 and shoot biomass ($R^2 = 0.485$) shows that plant growth creates consistent ecological gradients. Larger plants release more carbon below ground, providing root exudates that support microbial growth (Stockmann et al., 2013). This increased carbon flow favours copiotrophic taxa and may limit oligotrophic ones.

The modest PERMANOVA R^2 reflects the complex, high-dimensional nature of the data. The dataset includes 3,358 contigs, but biomass mainly influences 194 of them (5.8%). Much of the remaining variation likely comes from factors we did not measure, such as soil heterogeneity, microsite differences, random colonization, and historical effects.

Plant productivity influences microbial turnover through rhizosphere priming effects. Vigorous plant growth increases root exudation, forming nutrient-rich zones that stimulate microbial activity. Microbes grow quickly, exhaust resources, then die or are consumed, making space for new colonizers. High plant productivity shapes the rhizosphere environment by increasing moisture, nutrient levels, and energy availability, which selects for specific microbial groups. More productive plants maintain fast-changing, active communities, while less productive plants support more stable communities similar to bulk soil (Wang & Kuzyakov, 2024).

This pattern affects how metagenomic data should be interpreted. Standard diversity metrics may overlook real biological signals because they combine responses from many taxa that react in opposite ways. Analysing differential abundance at the feature level helps identify the specific microbes that respond to treatment. The empirical FDR method used here improves confidence in these results by accounting for data structure and zero inflation.

4.2.3 Copiotrophic taxa dominate high-biomass rhizospheres and support the living soil pathway

Actinobacteriota were the dominant phylum in high-biomass communities (41 contigs, 55%), with *Pseudonocardia* (10 contigs), *Geodermatophilus* (6), *Actinoplanes* (6), and *Blastococcus* (5) as key genera (Figure 3.5). This dominance fits well with their ecological roles. Members of Actinobacteriota produce secondary metabolites such as antibiotics, siderophores, and plant hormones that can promote plant growth and suppress pathogens. In particular, *Pseudonocardia* species are known mutualists that colonize plant surfaces and release antimicrobial compounds, indicating they likely act as plant growth-promoting bacteria in productive rhizospheres (Riahi et al., 2022).

Proteobacteria formed the second major group, with *Massilia* showing especially strong enrichment (four contigs, 0.78-2.4 Mb). *Massilia* species are recognized plant growth-promoting bacteria that colonize roots, produce auxins, solubilize phosphorus, and inhibit pathogens. Their genomes show diverse metabolic pathways for using plant compounds, producing antimicrobials, and synthesizing hormones (Dahal et al., 2021; C. Li et al., 2021; Zheng et al., 2017). The presence of several enriched *Massilia* genotypes suggests some level of functional redundancy or niche partitioning within rhizosphere communities.

These copiotrophic taxa share key traits that help them use abundant root exudates. They grow fast, respond quickly to available carbon, and outcompete slower organisms when resources are plentiful. This pattern matches broader ecological trends where high nutrient levels favour fast-growing groups like Proteobacteria and reduce oligotrophic taxa (Fierer et al., 2012).

The enrichment of nitrogen-cycling bacteria such as *Nitrosospira* and *Bradyrhizobium* shows

functional integration within the community. Root exudates support heterotrophic growth and provide carbon sources for nitrifiers. Higher nitrogen availability, in turn, promotes plant growth and strengthens positive feedback loops typical of the living soil pathway. This link between carbon and nitrogen dynamics aligns with the conceptual model where rhizosphere interactions drive both plant productivity and stable carbon formation through microbial turnover (Wang & Kuzyakov, 2024).

These compositional shifts directly support the living soil pathway framework. The rhizosphere enrichment of fast-growing Actinobacteriota and *Massilia* reflects active processing of root exudates as the primary carbon source in the living pathway. Their rapid turnover, indicated by high rRNA copy numbers and low GC content in some lineages, accelerates conversion of labile root-derived carbon into stable microbial necromass. This mechanism contrasts with the decomposition pathway, where bulk soil communities process more recalcitrant litter-derived substrates into loosely bound POM.

4.2.4 Oligotrophic specialists characterize low-biomass communities

Low-biomass soils harbored contrasting assemblages dominated by oligotrophic taxa: Acidobacteriota (5 contigs, 14%), Gemmatimonadota (6 contigs, 17%), and distinct Proteobacteria clades. Acidobacteriota are quintessential oligotrophs with slow growth, small cells, and streamlined genomes adapted to nutrient limitation. Two large Pyrinomonadaceae genomes (2.3-2.8 Mb) enriched at low biomass match their ecology: thriving in carbon-limited soils with efficient resource scavenging and stress tolerance. These organisms persist by employing high-affinity nutrient uptake, degrading recalcitrant compounds, and entering energy-efficient maintenance states.

Low-biomass soils contained different microbial groups dominated by oligotrophic taxa such as *Acidobacteriota* (5 contigs, 14%), *Gemmatimonadota* (6 contigs, 17%), and several distinct Proteobacteria clades (Figure 3.6). *Acidobacteriota* are classic oligotrophs with slow growth, small cells, and streamlined genomes suited to nutrient-poor conditions (Wong et al., 2024). Two large *Pyrinomonadaceae* genomes (2.3-2.8 Mb) enriched in low-biomass soils fit this pattern. They thrive in carbon-limited environments by using resources efficiently and

tolerating stress. These organisms survive through high-affinity nutrient uptake, degradation of complex compounds, and energy-saving maintenance states (Wüst et al., 2016).

Gemmatimonadota, common indicators of arid environments, comprise about 2% of global soil communities. Their enrichment here may reflect weaker rhizosphere influence, where low plant productivity and reduced root exudation create oligotrophic conditions that favor stress-tolerant specialists (Mujakić et al., 2022). Together with *Chloroflexota*, these taxa show flexible metabolism for using complex carbon sources unavailable to fast-growing copiotrophs (Petriglieri et al., 2023).

Differences within *Proteobacteria* are also informative. While *Massilia* (Gammaproteobacteria) dominated high-biomass soils, low-biomass communities included *Alphaproteobacteria* such as *Xanthobacteraceae* and *Reyranellaceae*, along with oligotrophic Burkholderiales (family SG8-39). This diversity within a single phylum shows that taxonomy alone does not fully explain ecological roles. The differential abundance analysis separated these groups clearly, highlighting the value of contig-level resolution for studying microbial ecology.

4.2.5 Microbial community shifts have direct implications for carbon stabilization

Our results support the living soil framework, which proposes that root–microbe interactions drive carbon stabilization. Microbial residues make up more than half of soil organic carbon in many ecosystems, showing that the microbial carbon pump plays a key role in long-term carbon storage (Liang et al., 2019).

The enrichment of plant growth-promoting bacteria such as *Pseudonocardia*, *Massilia*, and *Bradyrhizobium* points to mutual feedback between plants and microbes. These organisms enhance plant growth by producing auxins, solubilizing phosphorus, and suppressing pathogens. Larger plants, in turn, send more carbon below ground, releasing exudates that feed microbial growth. This feedback increases microbial turnover, which converts short-lived root exudates into stable, mineral-associated organic matter.

In high-biomass rhizospheres, steady inputs of labile carbon support large, fast-turnover

microbial communities whose death contributes persistent organic material to the soil. Rapid copiotrophic growth and decay increase the deposition of necromass into the soil matrix (Liang et al., 2019). Intense microbial activity also produces extracellular polymers that help form and stabilize aggregates. Within these aggregates, necromass becomes physically protected or chemically bound to minerals, which slows its breakdown.

Actinobacteriota are especially suited for carbon stabilization. Their genomes have high GC content (around 66%), which is linked to the production of amino sugars, the key components of necromass that resist decomposition (Miltner et al., 2012). As these bacteria die and lyse, their cell wall materials bind to minerals and can persist for long periods, from decades to centuries. The 41 enriched Actinobacteriota contigs identified here therefore represent not only active users of root exudates but also potential builders of stable soil carbon.

Contrasting oligotroph dominance in low-biomass communities suggests weaker rhizosphere influence. These soils likely receive less root-derived carbon, creating conditions where slow-growing specialists compete effectively. While contributing less necromass short-term due to lower biomass, they may exhibit higher carbon use efficiency, converting more available carbon into biomass rather than respired CO_2. Under the living soil framework, low-biomass soils would exhibit lower microbial turnover rates and reduced stable mineral-associated organic matter formation.

The dominance of oligotrophic taxa in low-biomass soils suggests a weaker rhizosphere effect. These soils likely receive less carbon from roots, creating environments where slow-growing specialists compete more successfully. Although these communities produce less necromass in the short term because of lower biomass, they may have higher carbon use efficiency, converting a greater share of available carbon into biomass rather than respired CO_2. Within the living soil framework, low-biomass soils would show slower microbial turnover and reduced formation of stable, mineral-associated organic matter.

This trend matches earlier findings showing that root-derived carbon is stabilized much more efficiently (76%) than litter-derived carbon (9%) (Berhongaray et al., 2019). The microbial community differences observed in this study may help explain these contrasting stabilization

rates. In productive rhizospheres, active plant-microbe interactions maintain steady microbial turnover, increase the accumulation of stable carbon, and influence nutrient cycling and soil structure through aggregation.

4.2.6 Circular genetic elements show independence from productivity gradients

The absence of biomass-related patterns in circular contigs contrasts with the chromosomal signals. We identified 448 circular contigs, but none showed significant changes in abundance with plant biomass. This suggests that horizontal gene transfer operates independently of the ecological gradients shaping main bacterial genomes (Smillie et al., 2010).

This decoupling likely reflects the different roles of plasmids and linear chromosomes. Plasmids usually carry accessory genes for traits such as antibiotic resistance, metal tolerance, or degradation of complex compounds, offering benefits that depend on local conditions rather than plant productivity (Nicholson et al., 2000). In contrast, chromosomal genes involved in core metabolism tend to respond directly to resource gradients. Plasmid-linked functions may instead react to other selective pressures, such as antibiotic production, metal exposure, or intermittent chemical inputs.

Circular element taxonomic composition differed greatly from that of chromosomal contigs. *Firmicutes* were notably more common (13%) even though they were almost absent among differentially abundant chromosomes. Members of this group form endospores that survive harsh conditions, and their plasmids may carry genes for sporulation that are less relevant in the rhizosphere (Nicholson et al., 2000). *Cyanobacteria* were also more frequent among circular elements despite being absent from biomass-associated linear contigs. These may represent dormant cells or specialized lineages with plasmids carrying photosynthetic genes active only under certain light conditions.

This pattern shows that microbes adapt through more than one mechanism. Community composition shifts mainly reflect resource availability, while horizontal gene transfer may respond to other selective forces. Long-read sequencing was especially valuable for identifying these circular elements, allowing recovery of 923 putatively complete replicons。

This is a major advantage since short-read assemblies rarely close circular genomes with confidence.

4.2.7 Methodological strengths and key limitations

Several methodological choices strengthened this analysis. First, empirical FDR calibration through permutation testing set data-driven significance thresholds that accounted for compositional effects and zero inflation. Second, filtering by both statistical significance ($p < \tau^*$) and effect size ($|\log_2 FC| \geq 0.5$) helped remove minor or uninformative differences (Li et al., 2022). Third, long-read assembly generated many large contigs (over 1 Mb) that represent near-complete genomes, allowing confident taxonomic and functional interpretation. The 11 large differentially abundant contigs (>1 Mb) are strong candidates for further annotation and possible cultivation.

However, some important limitations remain. First, metagenomic data measure DNA abundance rather than biological activity. Signals can include DNA from dormant cells or extracellular sources, which may obscure the microbes that are truly active. Second, contig-level testing presents challenges. Differences in contig length, completeness, and quality can influence abundance estimates. Large genomes split across several contigs may appear underrepresented, while conserved genes might recruit reads from multiple species. Strict mapping rules reduced cross-mapping, but some uncertainty remains for closely related strains.

Third, the biomass variable captures only one aspect of plant influence. Factors such as species identity, root structure, and exudate chemistry also shape microbial communities. We found no strong clustering by plant species, which may reflect the controlled experimental setup or real functional overlap among hosts.

Additionally, not all significant contigs were functionally annotated. Future studies should focus on gene-level analyses to identify metabolic pathways that link microbial activity with plant productivity. Fourth, soil datasets exhibit high dispersion and many zero counts. Our sample size of 26 may not be large enough to detect subtle shifts in rare taxa.

Finally, different differential abundance tools can produce varying results (Nearing et al., 2022; Yang & Chen, 2022). Although the edgeR approach used here is well established, other algorithms might highlight additional features. A combined strategy would improve confidence but was beyond the scope of this study.

4.3 Future directions

4.3.1 From assemblies to biological insight

The assembled contigs form a strong foundation, but their full value depends on detailed functional and taxonomic annotation. Standardized bacterial genome annotation would reveal complete metabolic pathways, regulatory networks, and mobile genetic elements across the 86 high-quality contigs (Schwengers et al., 2021). Integrating these genomes within the GTDB framework would classify the 34 currently unassigned contigs enriched in high-biomass soils, potentially uncovering novel lineages adapted to rhizosphere environments.

Targeted functional annotation of Actinobacteriota assemblies could address specific questions about their role in the living soil pathway (Blin et al., 2021). Characterizing biosynthetic gene clusters would identify their capacity to produce secondary metabolites such as antibiotics, siderophores, and signalling molecules that shape plant-microbe interactions. For the 923 circular replicons, plasmid-focused annotation would clarify their contribution to horizontal gene transfer and accessory functions, helping explain why these elements display ecological patterns distinct from chromosomal genes (McGuffie & Barrick, 2021).

Subsequent gene set enrichment analysis could then connect these annotations to ecological patterns (Korotkevich et al., 2021). Rather than focusing on individual genes, this method identifies entire pathways and modules associated with environmental variables, testing whether the observed taxonomic shifts represent true metabolic differentiation or simply reflect phylogenetic structure.

4.3.2 Cross-dataset comparative analysis

A major advantage of our genome-resolved approach is that these assemblies can serve as a unified custom database for analysing all three experiments in this thesis (temporal dynamics, plant species effects, soil type contrasts). Mapping reads from each dataset to the same reference would allow direct quantitative comparisons across treatment.

Principal component analysis of genome abundances across datasets could show whether the copiotrophic-oligotrophic gradient identified here is consistent across soil types, plant species, and sampling times. Do the same Actinobacteriota and *Massilia* strains respond to plant productivity in every context, or do soil type and plant identity select for different communities at similar productivity levels?

Differential abundance testing within comparable experimental blocks could address specific ecological questions. Do temporal dynamics follow predictable successions, with copiotrophs dominating early in the growing season when root exudation peaks and oligotrophs increasing later as exudation declines? Do plant species with contrasting root systems or exudate chemistries enrich distinct copiotrophic groups, or does high productivity always favour a similar set of microbes?

Cross-dataset integration would greatly strengthen ecological inference. Patterns that repeat across independent experiments are less likely to result from sampling noise and more likely to reflect real ecological processes. This unified database approach also maximizes the value of existing sequencing data, allowing new insights without additional experimental effort.

4.3.3 Longer-term experimental validation

While the bioinformatic analyses can begin right away, several longer-term studies would offer complementary insights. Targeted isolation and cultivation of dominant taxa such as *Pseudonocardia*, *Massilia*, and *Pyrinomonadaceae* would allow direct functional testing and inoculation experiments. Adding activity-based approaches like metatranscriptomics or metaproteomics would show which genes are actively expressed rather than simply present. Linking community composition to soil carbon fractionation through isotopic tracing, such as

^{13}C-labeled CO_2 experiments, would directly test predictions about the living soil pathway and microbial necromass contributions to stable carbon. These experiments would be highly valuable but require additional resources and fall beyond the immediate scope of computational analyses.

4.4 Conclusions

This thesis shows that long-read metagenomic approaches can recover near-complete genomes from complex soil communities, overcoming technical barriers that have limited soil microbiome research. The 86 contigs larger than 1 Mb and 923 circular replicons make it possible to link microbial identity with metabolic capacity and ecological function confidently.

Applying this framework to differential abundance analysis revealed that plant productivity structures soil microbial communities along a copiotrophic-oligotrophic gradient. High-biomass rhizospheres were dominated by fast-growing bacteria, especially Actinobacteriota (55% of mapped contigs) and *Massilia*, both adapted for rapid resource use. In contrast, low-biomass soils contained oligotrophic specialists such as *Acidobacteriota* and *Gemmatimonadota* suited to nutrient-poor conditions.

These community shifts have direct consequences for soil carbon cycling through the living soil pathway. The enrichment of copiotrophic taxa in high-biomass rhizospheres, together with their rapid turnover, likely accelerates the conversion of root exudates into stable microbial necromass. Actinobacteriota appear especially important in this process. Their high genomic GC content correlates with amino sugar production, yielding cell wall components that resist decomposition and persist as mineral-bound organic matter.

The findings also have practical implications for soil management. Encouraging vigorous plant growth through species selection, fertilization, or breeding could favour copiotrophic communities that enhance nutrient cycling and carbon sequestration. Because plant productivity and microbial structure are tightly connected, managing plant traits such as root architecture or exudate chemistry may help shape beneficial soil microbiomes.

Overall, this work highlights soil as a dynamic living system shaped by plant–microbe interactions. Community differences between rhizosphere and bulk soil, though occurring at micrometre scales, influence ecosystem processes such as carbon storage and nutrient availability. Understanding these links, from specific taxa to genome content to biogeochemical function, provides a foundation for using soil microbiomes to support sustainable agriculture and climate mitigation. The next steps include comprehensive functional annotation of assembled genomes and cross-dataset integration to test these patterns across contexts. Longer-term experimental validation and management trials will extend these computational foundations into applied outcomes.

Appendix

Supplementary Table 1 Summary statistics for the unified ONT metagenomic co-assembly.

Assembly Statistics	Assembly
# contigs (>= 1 kbp)	556967
# contigs (>= 10 kbp)	264089
# contigs (>= 20 kbp)	71839
# contigs (>= 50 kbp)	10888
# contigs (>= 1 Mbp)	86 (minimum coverage among these: 8.66x)
Largest contig (bp)	4516797
Avg. coverage depth	16
Coverage >= 1x (%)	99.81
Coverage >= 5x (%)	80.1
Coverage >= 10x (%)	54.66
# contigs (≥ 10 kbp OR ≥ 10x)	268290
# contigs (≥ 20 kbp OR ≥ 20x)	72061
# Circular contigs	923
# Circular contigs (≥ 10 kb OR ≥ 10x)	448 (minimum coverage among these: 1.01x)
# Circular contigs (≥ 20 kb OR ≥ 20x)	175 (minimum coverage among these: 1.56x)
Total bases of output contigs (Gbp)	7.691135072
Total bases of contigs > 10kb (Gbp)	5.634902355
Total bases of contigs > 20kb (Gbp)	3.043890337
Total bases of contigs > 10X (Gbp)	0.716887697
Total bases of contigs > 20X (Gbp)	0.141109321
Total bases of contigs > 10kb or > 10X (Gbp)	5.660392158
Total bases of contigs > 20kb or > 20X (Gbp)	3.045965642
Total bases of circular contigs (Mbp)	3.9854533
Total bases of circular contigs (> 10kb or > 10X) (Mbp)	3.6009224
Total bases of circular contigs (> 20kb or > 20X) (Mbp)	3.258957
Total bases of circular contigs (> 1Mb) (Mbp)	0.22357959
N50*	15514
N90	7668
auN*	91014
L50*	116258
L90	408780

GC (%)*	66.02
Reads mapping	
# mapped*	63916249
Mapped (%)	237.28
# singletons*	0
Singletons (%)	0
Per base quality	
# N's per 100 kbp*	0
# N's	0
Predicted genes	
# predicted rRNA genes	2311 + 333 part

* N50 is the contig length such that using longer or equal length contigs produces half (50%) of bases of the assembly. Similarly, Nx for any x between 0 and 100 could be computed (x% of assembly).

* L50 is the minimum number of contigs that produce half (50%) of the bases of the assembly. In other words, it's the number of contigs of length at least N50. Similarly, Lx for any x between 0 and 100 could be computed.

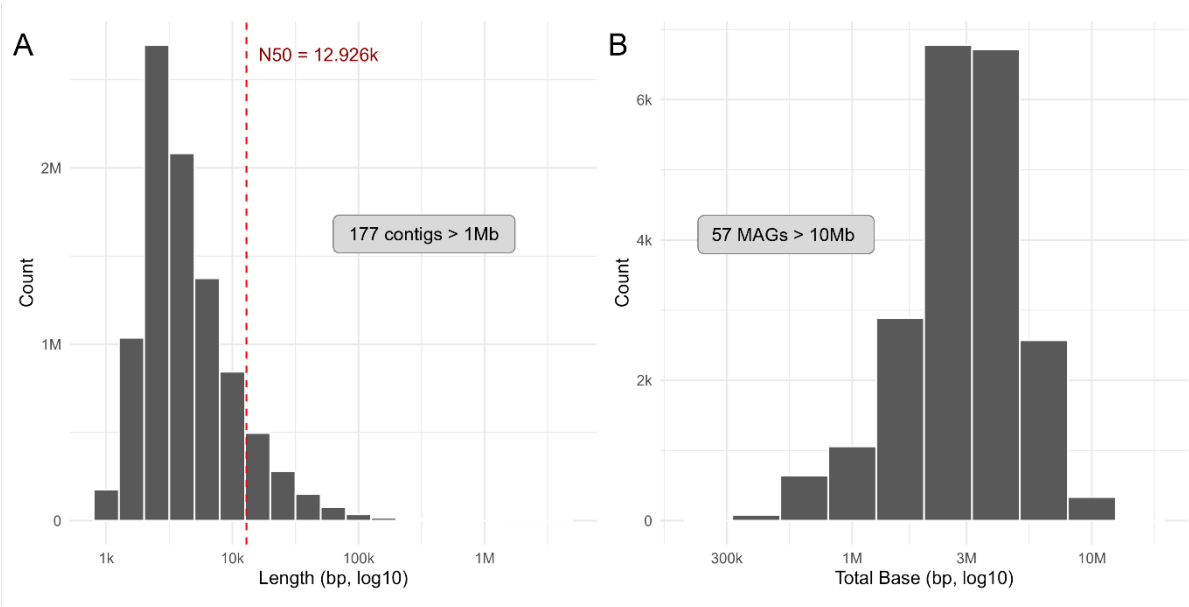
* anN is the area under the Nx curve.

* GC (%) is the total number of G and C nucleotides in the assembly, divided by the total length of the assembly.

* # mapped is the number of reads that mapped to the assembly. "Mapped % = 237.28" reflects multi-segment/ambiguous mappings common in metagenomes.

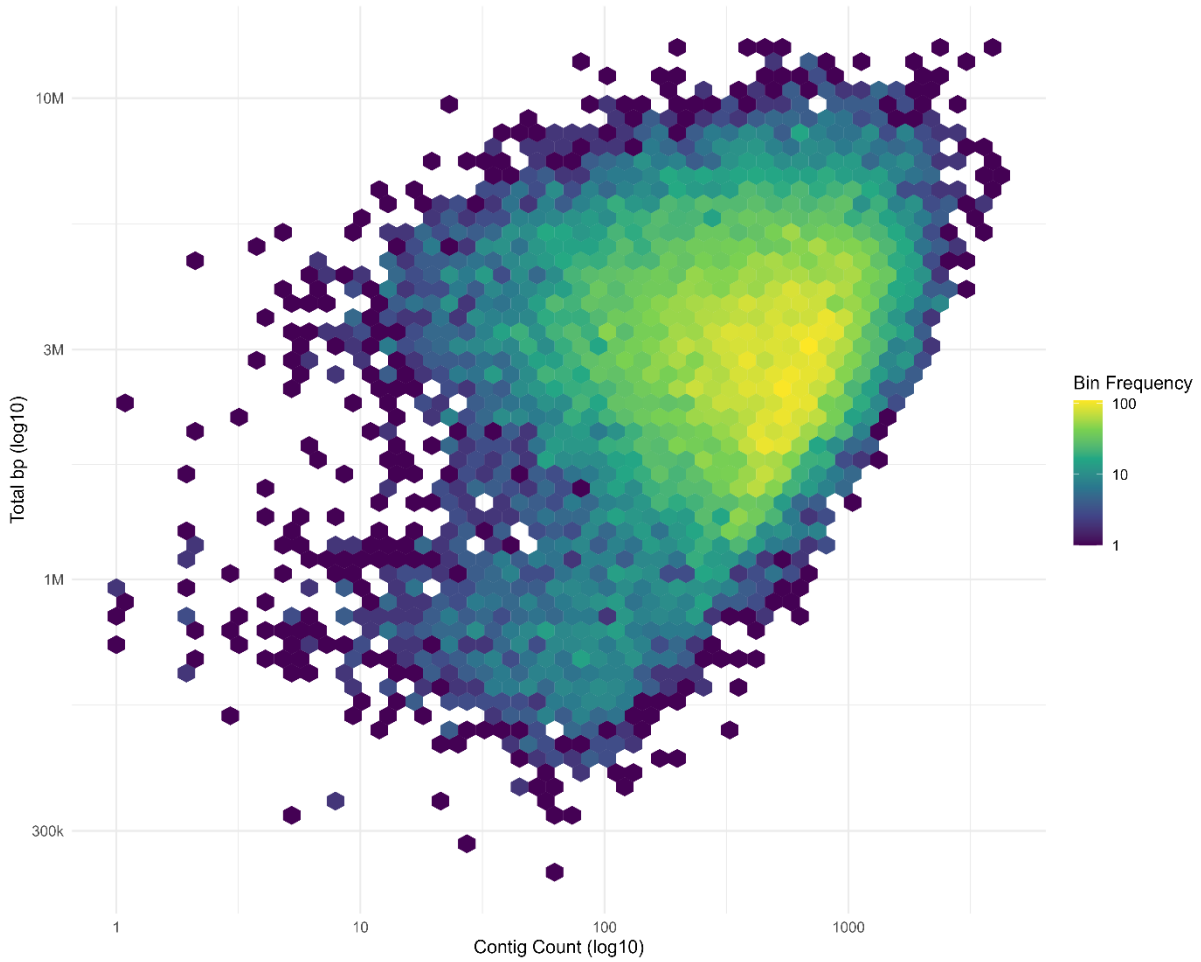
* # singletons is the number of reads which do not overlap any other reads.

* N's per 100 kbp is the average number of uncalled bases (N's) per 100,000 assembly bases.

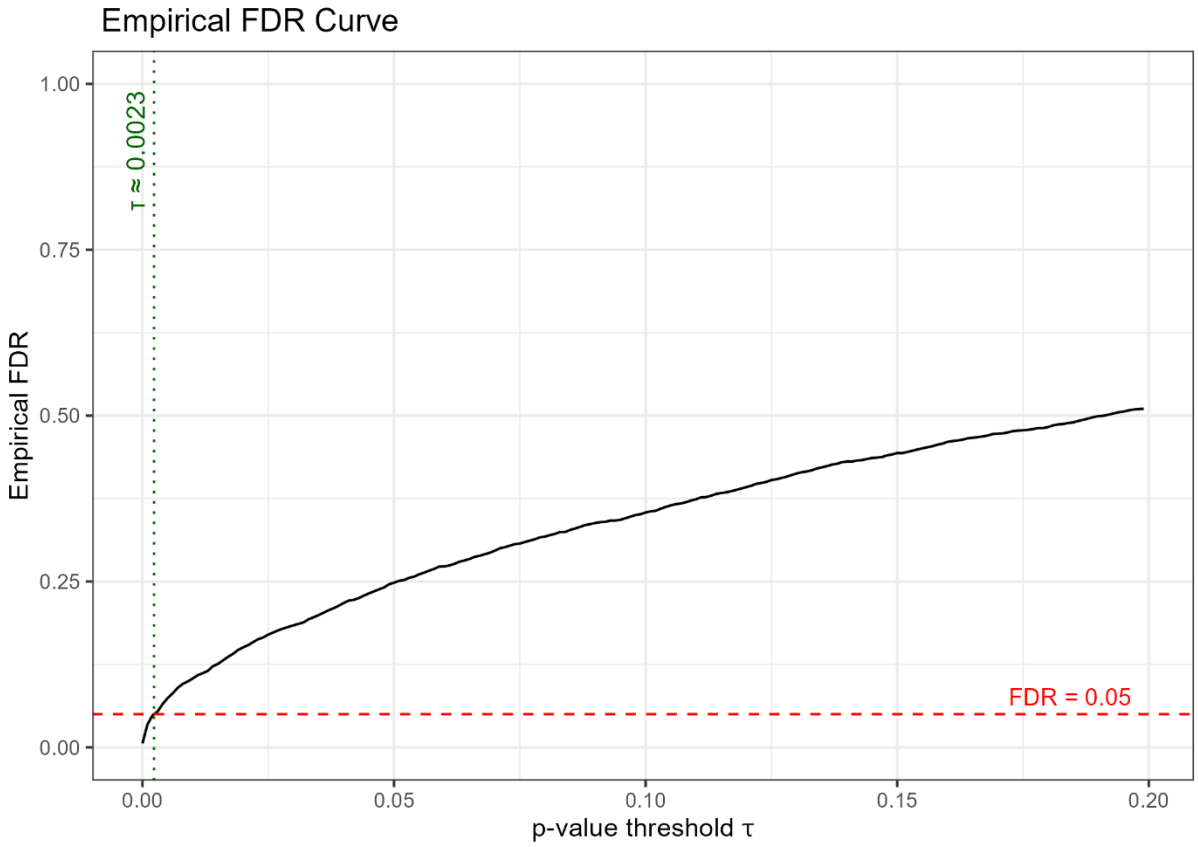


Supplementary Figure 1 (A) Contig length distribution in the SMAG catalogue. Histogram shows the distribution of contig lengths (log10 scale) across 9,280,391 assembled contigs from 3,304 global soil metagenomes. Red dashed line indicates N50 of 12.9 kb. The distribution is right-skewed with a peak around 2-3

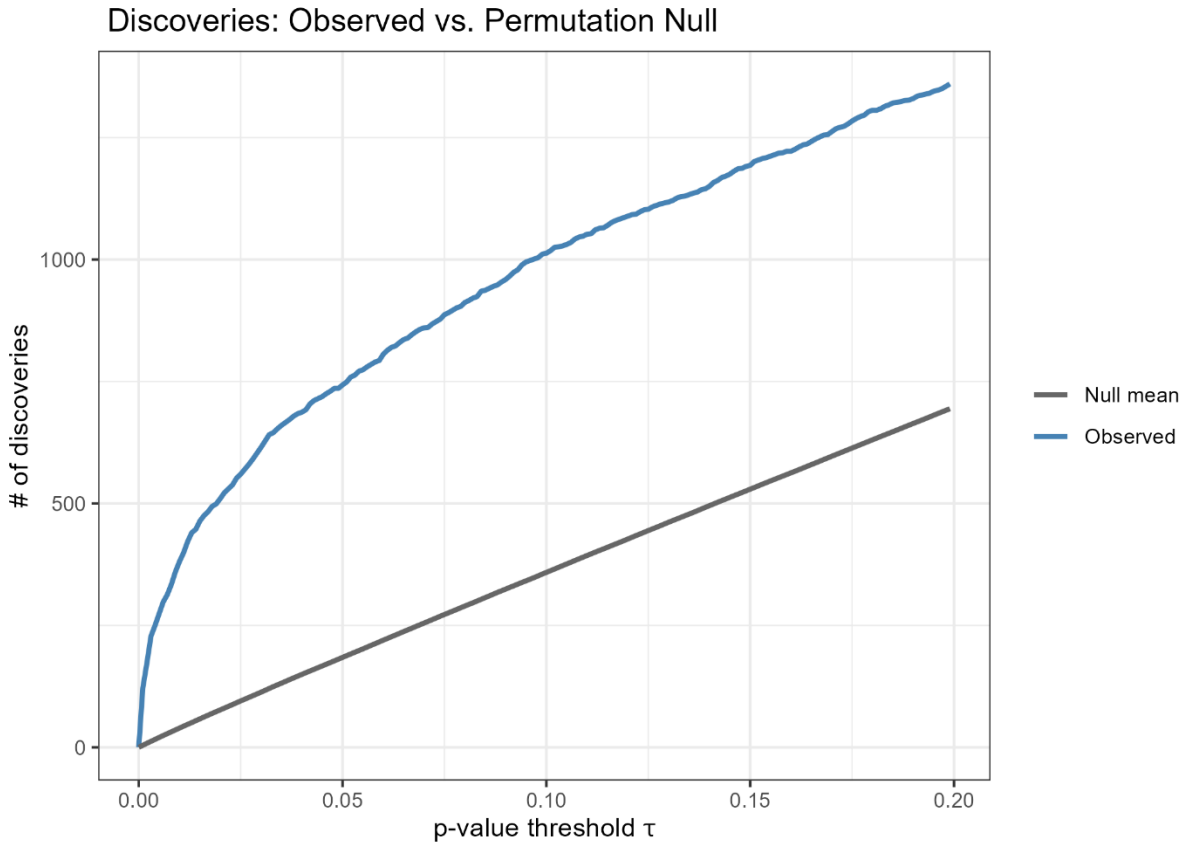
kb, reflecting typical fragmentation in short-read metagenomic assemblies. Despite overall fragmentation, 177 contigs exceed 1 Mb (annotated), demonstrating that short-read assemblies can occasionally achieve high contiguity for abundant, low-diversity organisms. **(B) Distribution of total assembly sizes for SMAG MAGs.** Histogram displays total genome sizes (log10 scale) for 21,077 species-level genome bins (SGBs) in the SMAG catalogue following quality filtering (⩾50% completeness). The distribution centres at 3 Mb with a secondary peak at 3.5 Mb, consistent with typical bacterial genome sizes. The narrow spread reflects successful filtering for near-complete genomes. Fifty-seven MAGs exceed 10 Mb (annotated), likely representing large prokaryotic genomes or potential eukaryotic contamination.



Supplementary Figure 2 Relationship between MAG contig count and total genome size. Hex-bin density plot shows the joint distribution of contig count (log10 scale) and total assembled length (log10 scale) for 21,077 SMAG species-level bins. Colour intensity represents bin frequency. The positive correlation (diagonal pattern) reflects the expected relationship where larger genomes contain more contigs. The dense core (yellow) around 100-1000 contigs and 3-10 Mb represents typical medium-to-high-quality MAGs. Upper-left outliers (few contigs, large size) represent exceptionally well-assembled MAGs, while right-side outliers (many contigs, moderate size) represent fragmented but complete genomes.



Supplementary Figure 3 Empirical FDR curve. Empirical false discovery rate (FDR) plotted against p-value threshold (τ). The curve was generated by comparing the number of significant contigs in the observed data to the mean number in 500 permutations where shoot mass labels were randomly shuffled. The vertical dotted line indicates $\tau^* = 0.0023$, where empirical FDR ≈ 0.05 (horizontal red dashed line).



Supplementary Figure 4 Discovery counts: observed versus permutation null. Number of contigs declared significant ($p \leq \tau$) plotted against p-value threshold (τ) for the observed data (blue line) and the mean of 500 permutations (grey line). The consistent separation between observed and null discovery curves demonstrates genuine biological associations with plant shoot biomass. At $\tau^* = 0.0023$, approximately 194 contigs were significant in the observed data versus ~10 expected under the null, yielding an empirical FDR of 0.05.

Supplementary Table 2 Top 20 differentially abundant contigs with taxonomic annotations

contig_id	MAG_ID	FDR	direction	length	coverage	circular	phylum	class	order	family	genus
ctg169286	SRR9650533rbin.13	0.002036313	enriched_in_large	106757	17.01	no	p__Actinobacteriota	c__Actinomycetia	o__Mycobacteriales	f__Pseudonocardiaceae	g__Pseudonocardia
ctg160376	SRR3954448rbin.13	0.002036313	enriched_in_large	361377	8.76	no	p__Actinobacteriota	c__Actinomycetia	o__Propionibacteriales	f__Kribbellaceae	g__Kribbella
ctg54092	SRR8438987rbin.18	0.002651754	enriched_in_large	334630	17.14	no	p__Actinobacteriota	c__Actinomycetia	o__Mycobacteriales	f__Micromonosporaceae	g__
ctg652443	SRR9691051rbin.24	0.006580167	enriched_in_large	890334	14.35	no	p__Actinobacteriota	c__Actinomycetia	o__Mycobacteriales	f__Micromonosporaceae	g__Actinoplanes
ctg435168	SRR5195146rbin.11	0.002036313	enriched_in_large	352044	17.78	no	p__Chloroflexota	c__Ellin6529	o__QHBO01	f__QHBO01	g__
ctg463047	SRR5195146rbin.11	0.002651754	enriched_in_large	158074	15.47	no	p__Chloroflexota	c__Ellin6529	o__QHBO01	f__QHBO01	g__
ctg586707	SRR5195146rbin.11	0.005722642	enriched_in_large	711416	16.88	no	p__Chloroflexota	c__Ellin6529	o__QHBO01	f__QHBO01	g__
ctg608997	S140bin.16	0.005583258	enriched_in_small	368615	8.33	no	p__Gemmatimonadota	c__Gemmatimonadetes	o__Gemmatimonadales	f__GWC2-71-9	g__JACDDX01
ctg294999	SRR6517655rbin.2	0.003902304	enriched_in_small	130586	11.43	no	p__Myxococcota	c__Myxococcia	o__Myxococcales	f__Myxococcaceae	g__Archangium
ctg349999	SRR7448326rbin.5	0.002036313	enriched_in_large	782969	24.16	no	p__Proteobacteria	c__Gammaproteobacteria	o__Burkholderiales	f__Burkholderiaceae	g__Massilia
ctg195233	SRR7448326rbin.5	0.002036313	enriched_in_large	2409072	21.89	no	p__Proteobacteria	c__Gammaproteobacteria	o__Burkholderiales	f__Burkholderiaceae	g__Massilia

contig_id	MAG_ID	FDR	direction	length	coverage	circular	phylum	class	order	family	genus
ctg317589	SRR7448326rbin.5	0.002651754	enriched_in_large	2062936	24.85	no	p__Proteobacteria	c__Gammaproteobacteria	o__Burkholderiales	f__Burkholderiaceae	g__Massilia
ctg446438	SRR5133179rbin.1	0.002651754	enriched_in_small	873416	15.33	no	p__Proteobacteria	c__Gammaproteobacteria	o__Burkholderiales	f__Burkholderiaceae	g__LX47W
ctg320552	SRR8487026rbin.1	0.004086925	enriched_in_small	279368	10.59	no	p__Proteobacteria	c__Alphaproteobacteria	o__Rhizobiales	f__Xanthobacteraceae	g__BOG-931
ctg293271	SRR8487026rbin.1	0.006580167	enriched_in_small	112548	10.19	no	p__Proteobacteria	c__Alphaproteobacteria	o__Rhizobiales	f__Xanthobacteraceae	g__BOG-931

Supplementary Table 3 Highly differentially abundant contigs from the heatmap with taxonomic annotations

contig_id	MAG_ID	FDR	direction	length	coverage	circular	classification	phylum	class	order	family	genus
ctg608997	S140bin.16	0.005583258	enriched_in_small	368615	8.33	no	d__Bacteria	p__Gemmatimonadota	c__Gemmatimonadetes	o__Gemmatimonadales	f__GWC2-71-9	g__JACDDX01
ctg446438	SRR5133179rbin.1	0.002651754	enriched_in_small	873416	15.33	no	d__Bacteria	p__Proteobacteria	c__Gammaproteobacteria	o__Burkholderiales	f__Burkholderiaceae	g__LX47W
ctg212521	NA	0.002036313	enriched_in_small	79279	5.51	no	NA	NA	NA	NA	NA	NA
ctg453700	SRR5215207rbin.29	0.008440778	enriched_in_small	256340	8.76	no	d__Bacteria	p__Proteobacteria	c__Alphaproteobacteria	o__Reyranellales	f__Reyranellaceae	g__Reyranella
ctg526371	SRR3982071rbin.1	0.007072599	enriched_in_small	2809112	14.71	no	d__Bacteria	p__Acidobacteriota	c__Blastocatellia	o__Pyrinomonadales	f__Pyrinomonadaceae	g__PSRF01

contig_id	MAG_ID	FDR	direction	length	coverage	circular	classification	phylum	class	order	family	genus
ctg383582	SRR5195141rbin.6	0.007509043	enriched_in_small	2311199	15.21	no	d__Bacteria	p__Acidobacteriota	c__Blastocatellia	o__Pyrinomonadales	f__Pyrinomonadaceae	g__PSRF01
ctg349999	SRR7448326rbin.5	0.002036313	enriched_in_large	782969	24.16	no	d__Bacteria	p__Proteobacteria	c__Gammaproteobacteria	o__Burkholderiales	f__Burkholderiaceae	g__Massilia
ctg641799	SRR7448326rbin.5	0.006580167	enriched_in_large	1196363	20.61	no	d__Bacteria	p__Proteobacteria	c__Gammaproteobacteria	o__Burkholderiales	f__Burkholderiaceae	g__Massilia
ctg195233	SRR7448326rbin.5	0.002036313	enriched_in_large	2409072	21.89	no	d__Bacteria	p__Proteobacteria	c__Gammaproteobacteria	o__Burkholderiales	f__Burkholderiaceae	g__Massilia
ctg317589	SRR7448326rbin.5	0.002651754	enriched_in_large	2062936	24.85	no	d__Bacteria	p__Proteobacteria	c__Gammaproteobacteria	o__Burkholderiales	f__Burkholderiaceae	g__Massilia

Supplementary Table 4 Long differentially abundant contigs (>1 Mb) with taxonomic annotations

contig_id	MAG_ID	FDR	direction	length	coverage	circular	classification	phylum	class	order	family	genus
ctg356117	SRR5919202rbin.1	0.0136 66998	enriched_in_large	3267749	34.44	no	d__Bacteria	p__Actinobacteriota	c__Actinomycetia	o__Mycobacteriales	f__Geodermatophilaceae	g__Geodermatophilus
ctg195233	SRR7448326rbin.5	0.0020 36313	enriched_in_large	2409072	21.89	no	d__Bacteria	p__Proteobacteria	c__Gammaproteobacteria	o__Burkholderiales	f__Burkholderiaceae	g__Massilia
ctg158538	SRR9650533rbin.11	0.0084 40778	enriched_in_large	2088099	34.31	no	d__Bacteria	p__Actinobacteriota	c__Actinomycetia	o__Mycobacteriales	f__Geodermatophilaceae	g__Geodermatophilus
ctg317589	SRR7448326rbin.5	0.0026 51754	enriched_in_large	2062936	24.85	no	d__Bacteria	p__Proteobacteria	c__Gammaproteobacteria	o__Burkholderiales	f__Burkholderiaceae	g__Massilia

contig_id	MAG_ID	FDR	direction	length	coverage	circular	classification	phylum	class	order	family	genus
ctg336357	SRR5133179rbin.1	0.017177822	enriched_in_large	1740693	13.71	no	d__Bacteria	p__Proteobacteria	c__Gammaproteobacteria	o__Burkholderiales	f__Burkholderiaceae	g__LX47W
ctg641799	SRR7448326rbin.5	0.006580167	enriched_in_large	1196363	20.61	no	d__Bacteria	p__Proteobacteria	c__Gammaproteobacteria	o__Burkholderiales	f__Burkholderiaceae	g__Massilia
ctg526371	SRR3982071rbin.1	0.007072599	enriched_in_small	2809112	14.71	no	d__Bacteria	p__Acidobacteriota	c__Blastocatellia	o__Pyrinomonadales	f__Pyrinomonadaceae	g__PSRF01
ctg383582	SRR5195141rbin.6	0.007509043	enriched_in_small	2311199	15.21	no	d__Bacteria	p__Acidobacteriota	c__Blastocatellia	o__Pyrinomonadales	f__Pyrinomonadaceae	g__PSRF01
ctg631138	SRR8185410rbin.2	0.030361481	enriched_in_small	2106942	17.31	no	d__Bacteria	p__Proteobacteria	c__Gammaproteobacteria	o__Burkholderiales	f__SG8-39	g__SCGC-AG-212-J23
ctg462036	NA	0.025600736	enriched_in_small	1026809	13.09	no	NA	NA	NA	NA	NA	NA
ctg242959	SRR12170509bin.29	0.037099688	enriched_in_small	1011816	18.53	no	d__Bacteria	p__Proteobacteria	c__Gammaproteobacteria	o__Burkholderiales	f__SG8-39	g__SCGC-AG-212-J23

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